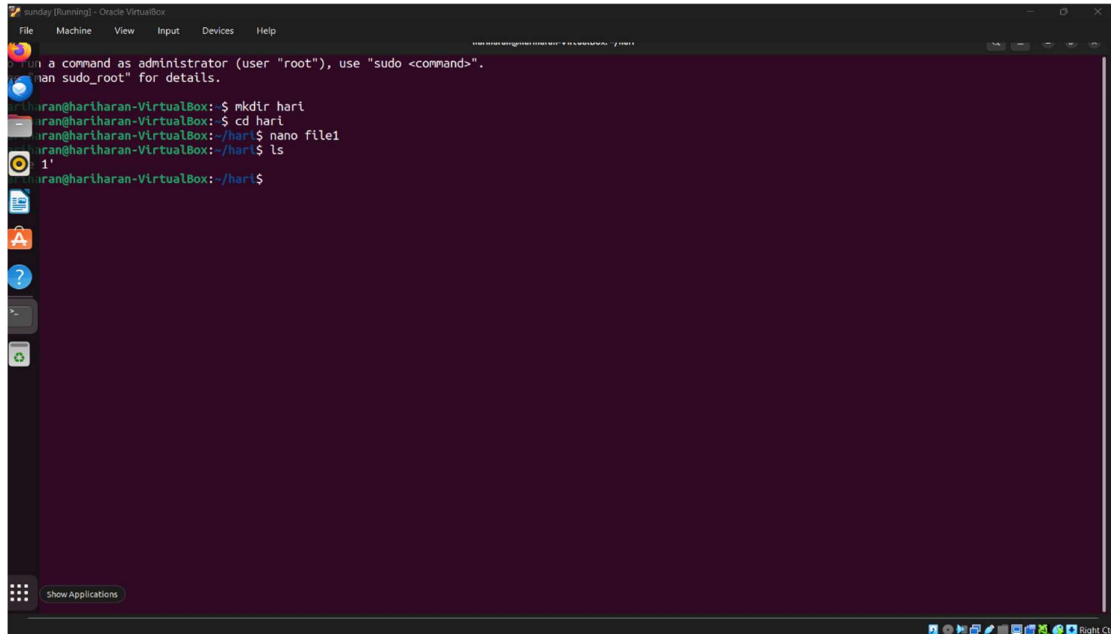


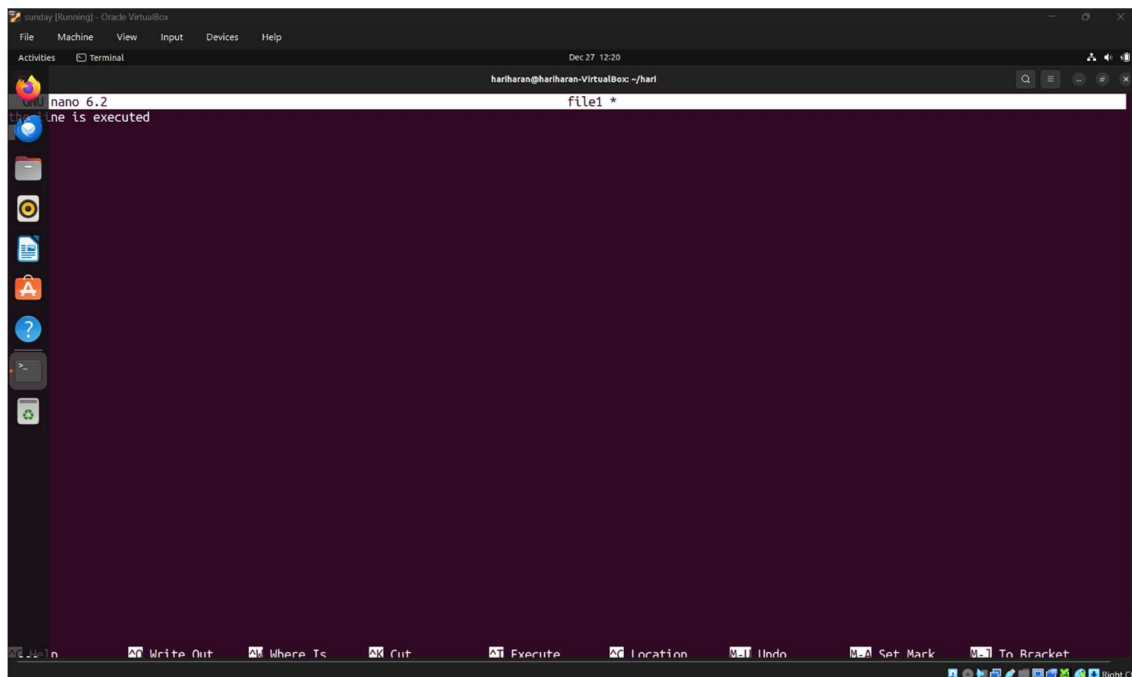
LINUX USER NAVIGATION COMMANDS

1. Make Directory with the help of command “mkdir” and change directory using “cd” and create an file using the command “nano”.



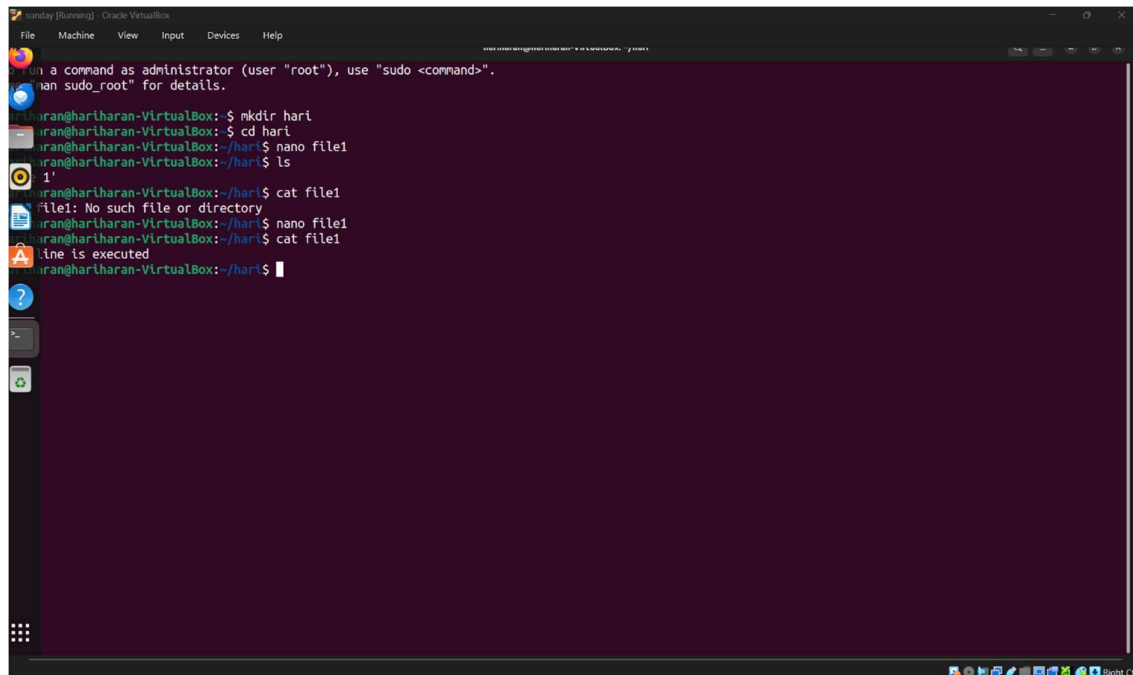
```
haran@hartharan-VirtualBox:~$ sudo su
haran@hartharan-VirtualBox:~$ mkdir hari
haran@hartharan-VirtualBox:~$ cd hari
haran@hartharan-VirtualBox:~/hari$ nano file1
haran@hartharan-VirtualBox:~/hari$ ls
file1
haran@hartharan-VirtualBox:~/hari$
```

Using “nano” command we create any type of file and save it.



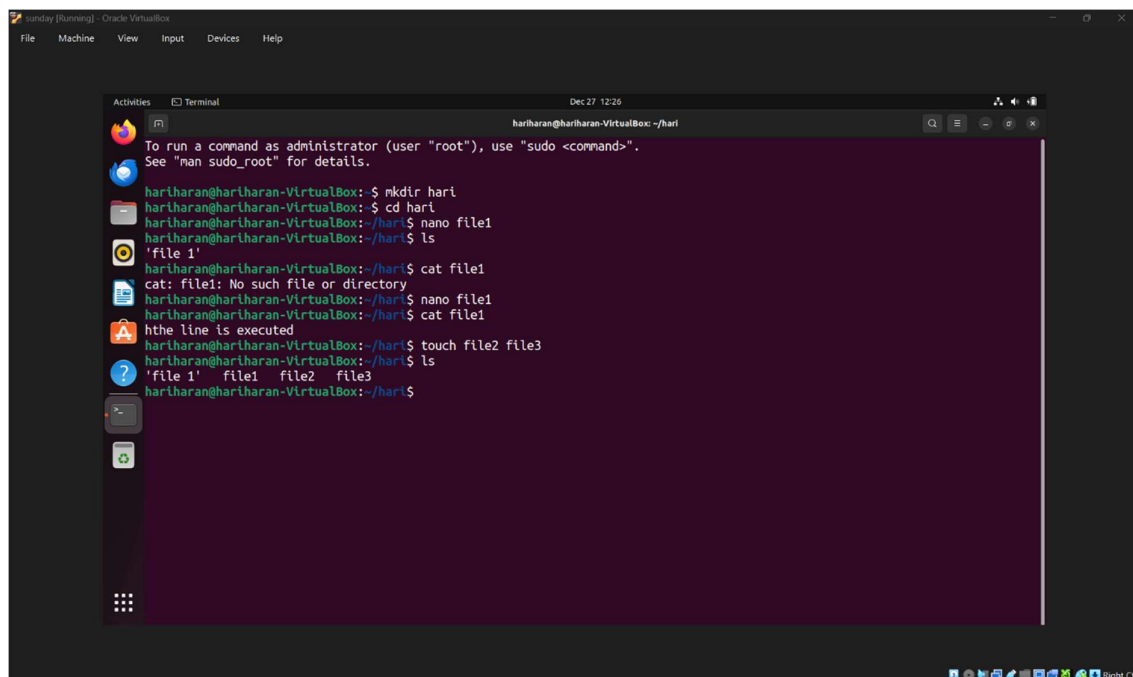
```
nano 6.2
file1 *
^O Write Out
^W Where Is
^X Cut
^E Execute
^L Location
^U Undo
^M Set Mark
^T To Bracket
```

Using “cat” command to view the inside the file.



```
haranghartharan-VirtualBox: ~$ mkdir hari
haranghartharan-VirtualBox: ~$ cd hari
haranghartharan-VirtualBox: ~/hari$ nano file1
haranghartharan-VirtualBox: ~/hari$ ls
file1
haranghartharan-VirtualBox: ~/hari$ cat file1
cat: file1: No such file or directory
haranghartharan-VirtualBox: ~/hari$ nano file1
haranghartharan-VirtualBox: ~/hari$ cat file1
line is executed
haranghartharan-VirtualBox: ~/hari$
```

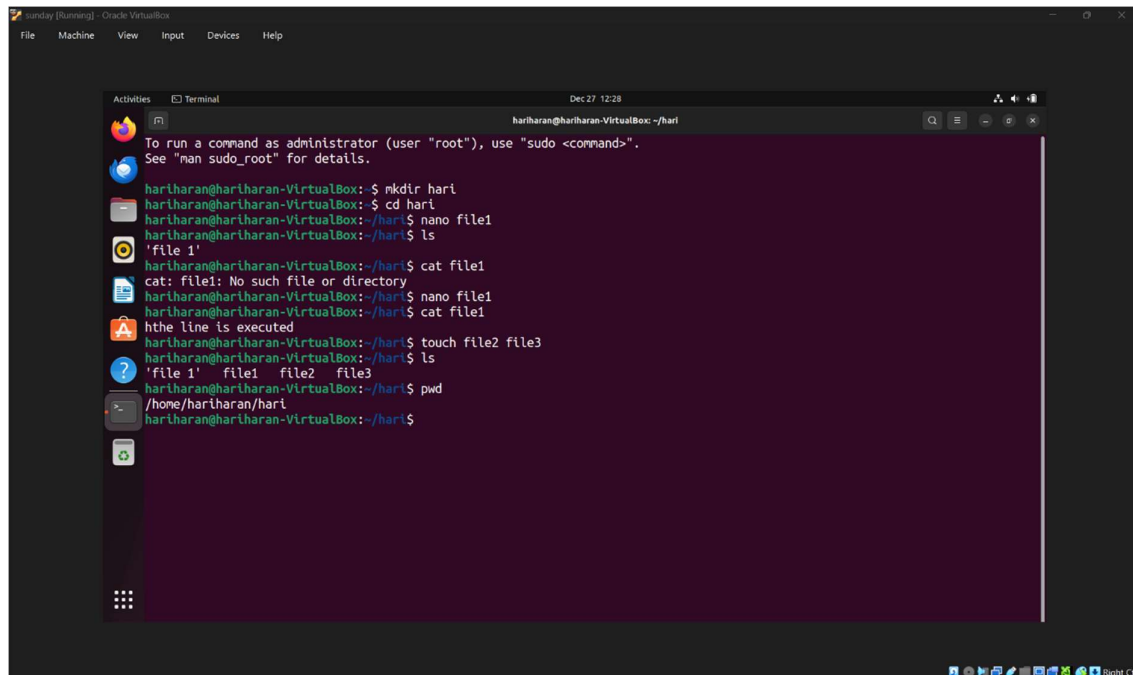
Using “touch” command to create a file.



```
hartharan@hartharan-VirtualBox: ~/hari
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

hartharan@hartharan-VirtualBox: ~$ mkdir hari
hartharan@hartharan-VirtualBox: ~$ cd hari
hartharan@hartharan-VirtualBox: ~/hari$ nano file1
hartharan@hartharan-VirtualBox: ~/hari$ ls
file1
hartharan@hartharan-VirtualBox: ~/hari$ cat file1
cat: file1: No such file or directory
hartharan@hartharan-VirtualBox: ~/hari$ nano file1
hartharan@hartharan-VirtualBox: ~/hari$ cat file1
the line is executed
hartharan@hartharan-VirtualBox: ~/hari$ touch file2 file3
hartharan@hartharan-VirtualBox: ~/hari$ ls
file1 file2 file3
hartharan@hartharan-VirtualBox: ~/hari$
```

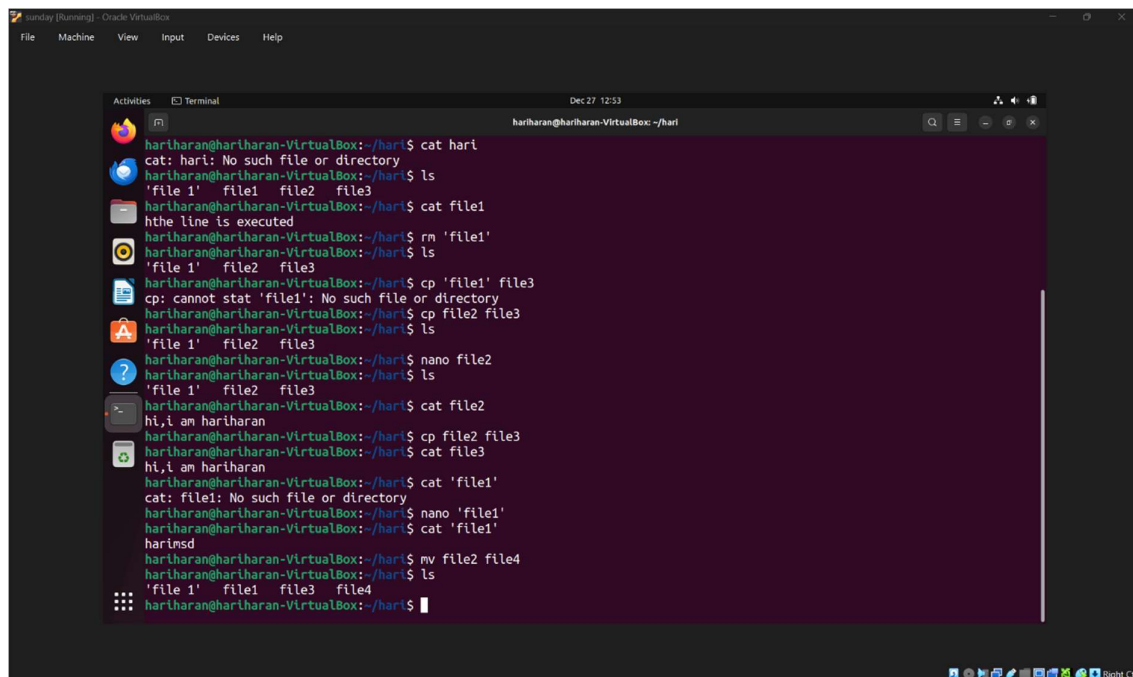
Using “pwd” command print working directory.



The screenshot shows a terminal window titled "hariharang@hariharan-VirtualBox: ~/hari". The user has executed several commands to create and manage files:

```
hariharang@hariharan-VirtualBox:~$ mkdir hari
hariharang@hariharan-VirtualBox:~/hari$ cd hari
hariharang@hariharan-VirtualBox:~/hari$ nano file1
'file 1'
hariharang@hariharan-VirtualBox:~/hari$ cat file1
cat: file1: No such file or directory
hariharang@hariharan-VirtualBox:~/hari$ nano file1
hariharang@hariharan-VirtualBox:~/hari$ cat file1
hthe line is executed
hariharang@hariharan-VirtualBox:~/hari$ touch file2 file3
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file1  file2  file3
hariharang@hariharan-VirtualBox:~/hari$ pwd
/home/hariharan/hari
hariharang@hariharan-VirtualBox:~/hari$
```

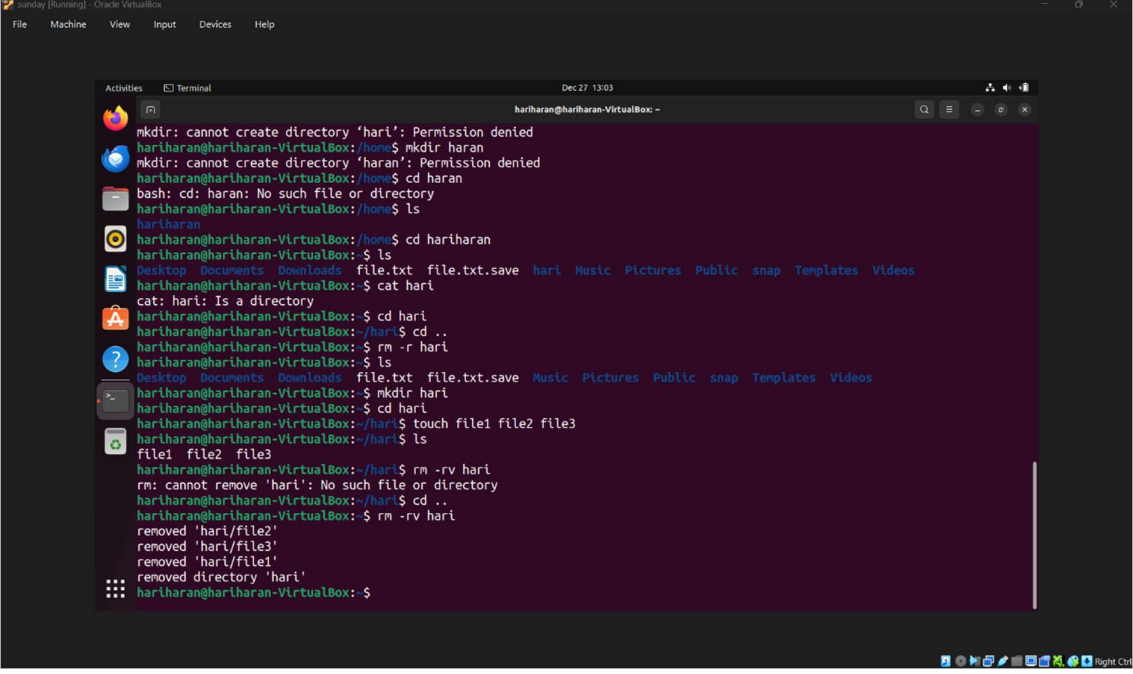
Using “cp” command to copy one file to another file and using “mv” command to move one file to another file.



The screenshot shows a terminal window titled "hariharang@hariharan-VirtualBox: ~/hari". The user has executed several commands to copy and move files:

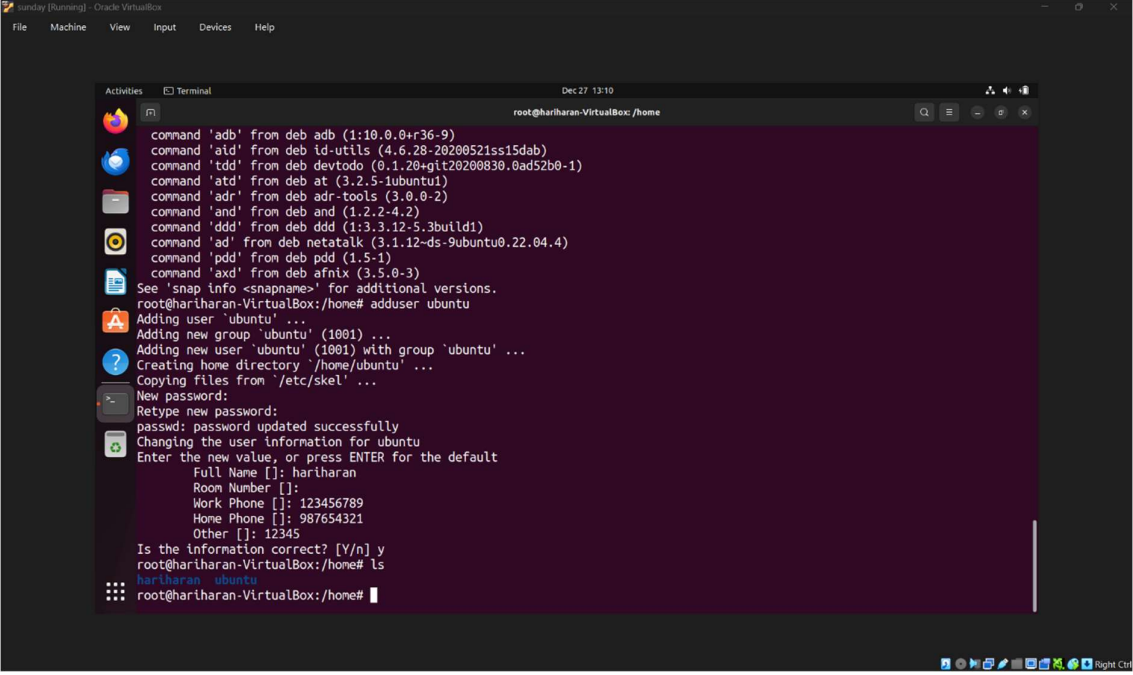
```
hariharang@hariharan-VirtualBox:~/hari$ cat hari
cat: hari: No such file or directory
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file1  file2  file3
hariharang@hariharan-VirtualBox:~/hari$ cat file1
hthe line is executed
hariharang@hariharan-VirtualBox:~/hari$ rm 'file1'
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file2  file3
hariharang@hariharan-VirtualBox:~/hari$ cp 'file1' file3
cp: cannot stat 'file1': No such file or directory
hariharang@hariharan-VirtualBox:~/hari$ cp file2 file3
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file2  file3
hariharang@hariharan-VirtualBox:~/hari$ nano file2
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file2  file3
hariharang@hariharan-VirtualBox:~/hari$ cat file2
hi,i an hariharan
hariharang@hariharan-VirtualBox:~/hari$ cp file2 file3
hariharang@hariharan-VirtualBox:~/hari$ cat file3
hi,i an hariharan
hariharang@hariharan-VirtualBox:~/hari$ cat 'file1'
cat: file1: No such file or directory
hariharang@hariharan-VirtualBox:~/hari$ nano 'file1'
hariharang@hariharan-VirtualBox:~/hari$ cat 'file1'
harimsd
hariharang@hariharan-VirtualBox:~/hari$ mv file2 file4
hariharang@hariharan-VirtualBox:~/hari$ ls
'file 1'  file1  file3  file4
hariharang@hariharan-VirtualBox:~/hari$
```

Using “rm -rv” to remove file and directory.



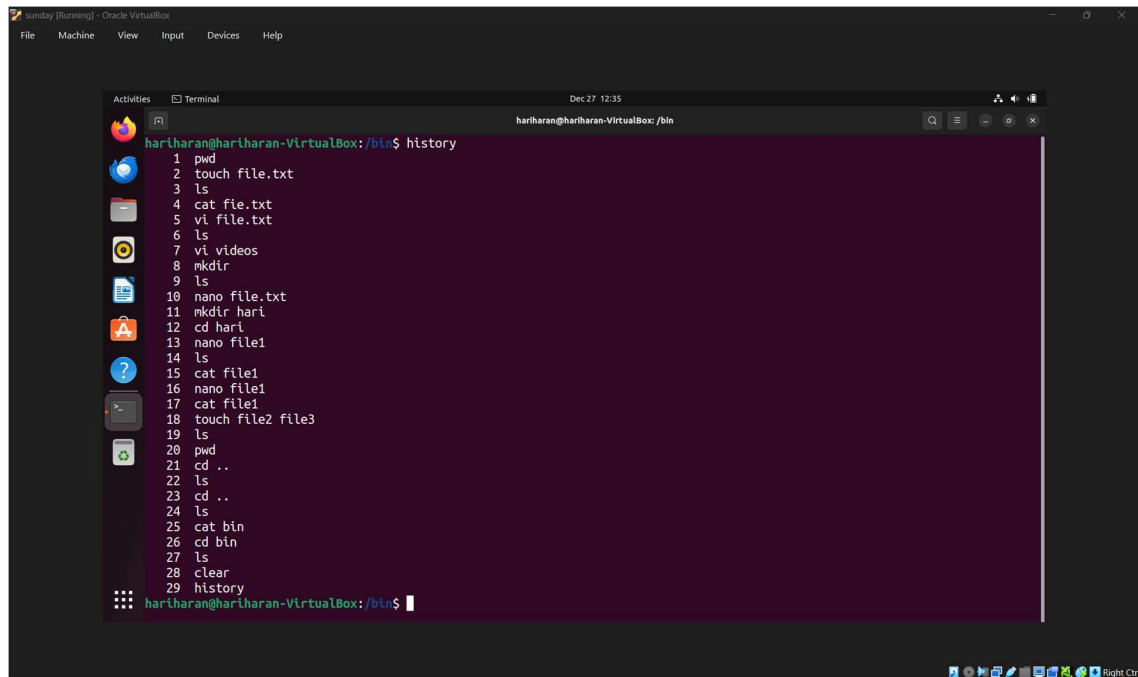
```
hariharan@hariharan-VirtualBox: ~$ mkdir haran
mkdir: cannot create directory 'haran': Permission denied
hariharan@hariharan-VirtualBox: ~$ cd haran
bash: cd: haran: No such file or directory
hariharan@hariharan-VirtualBox: ~$ ls
Desktop Documents Downloads file.txt file.txt.save hari Music Pictures Public snap Templates Videos
hariharan@hariharan-VirtualBox: ~$ cd hariharan
hariharan@hariharan-VirtualBox: ~$ ls
Desktop Documents Downloads file.txt file.txt.save hari Music Pictures Public snap Templates Videos
hariharan@hariharan-VirtualBox: ~$ cat hari
cat: hari: Is a directory
hariharan@hariharan-VirtualBox: ~$ cd hari
hariharan@hariharan-VirtualBox: ~/hari$ cd ..
hariharan@hariharan-VirtualBox: ~$ rm -r hari
hariharan@hariharan-VirtualBox: ~$ ls
Desktop Documents Downloads file.txt file.txt.save Music Pictures Public snap Templates Videos
hariharan@hariharan-VirtualBox: ~$ mkdir hari
hariharan@hariharan-VirtualBox: ~$ cd hari
hariharan@hariharan-VirtualBox: ~/hari$ touch file1 file2 file3
hariharan@hariharan-VirtualBox: ~/hari$ ls
file1 file2 file3
hariharan@hariharan-VirtualBox: ~/hari$ rm -rv hari
rm: cannot remove 'hari': No such file or directory
hariharan@hariharan-VirtualBox: ~/hari$ cd ..
hariharan@hariharan-VirtualBox: ~$ rm -rv hari
removed 'hari/file2'
removed 'hari/file3'
removed 'hari/file1'
removed directory 'hari'
hariharan@hariharan-VirtualBox: ~$
```

Using “sudu su” command to add user and “ls” command to list the user.



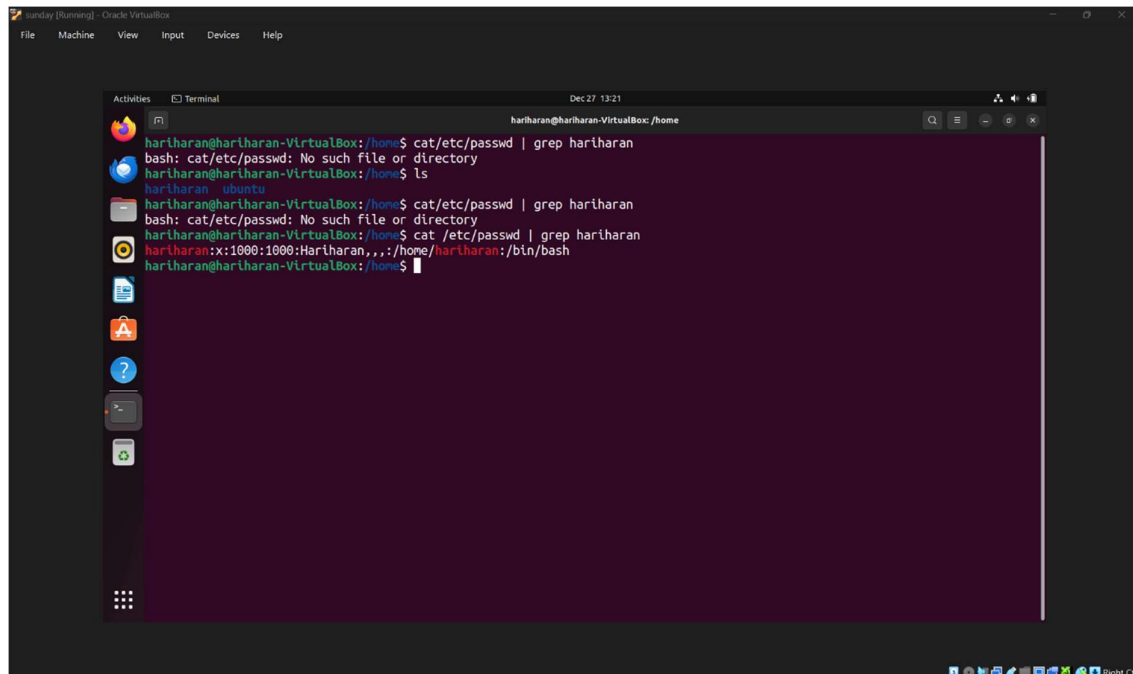
```
root@hariharan-VirtualBox: /home# command 'adb' from deb adb (1:10.0.0+r36-9)
command 'aid' from deb id-utils (4.6.28-20200521ss15dab)
command 'tdd' from deb devtodo (0.1.20+git20200830.0ad52b0-1)
command 'atd' from deb at (3.2.5-1ubuntu1)
command 'adr' from deb adr-tools (3.0.0-2)
command 'and' from deb and (1.2.2-4.2)
command 'ddd' from deb ddd (1:3.3.12-5.3build1)
command 'ad' from deb netatalk (3.1.12-ds-9ubuntu0.22.04.4)
command 'pdd' from deb pdd (1.5-1)
command 'axd' from deb afnix (3.5.0-3)
See 'snap info <snapnames>' for additional versions.
root@hariharan-VirtualBox: /home# adduser ubuntu
Adding user 'ubuntu' ...
Adding new group 'ubuntu' (1001) ...
Adding new user 'ubuntu' (1001) with group 'ubuntu' ...
Creating home directory '/home/ubuntu' ...
Copying files from '/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ubuntu
Enter the new value, or press ENTER for the default
Full Name []: hariharan
Room Number []:
Work Phone []: 123456789
Home Phone []: 987654321
Other []: 12345
Is the information correct? [Y/n] y
root@hariharan-VirtualBox: /home# ls
hariharan ubuntu
root@hariharan-VirtualBox: /home#
```

History of this using command.



LINUX ADMINISTRATOR COMMANDS AND FILE PERMISSION COMMANDS

- Using “Cat/etc/passwd | grep” command to identify the single user.



The screenshot shows a terminal window titled "hartharan@hartharan-VirtualBox: /home". The terminal displays the following commands and output:

```
hartharan@hartharan-VirtualBox:/home$ cat/etc/passwd | grep hartharan
bash: cat/etc/passwd: No such file or directory
hartharan@hartharan-VirtualBox:/home$ ls
hartharan  ubuntu
hartharan@hartharan-VirtualBox:/home$ cat/etc/passwd | grep hartharan
bash: cat/etc/passwd: No such file or directory
hartharan@hartharan-VirtualBox:/home$ cat /etc/passwd | grep hartharan
hartharan:x:1000:1000:Hartharan,,,:/home/hartharan:/bin/bash
hartharan@hartharan-VirtualBox:/home$
```

- Using “Cat/etc/passwd to identify the all user.

```
hariharan@hariharan-VirtualBox: /home$ cat/etc/passwd | grep hariharan
hariharan:x:1000:1000:Hariharan,,,:/home/hariharan:/bin/bash
hariharan@hariharan-VirtualBox: /home$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mail List Manager:/var/list:/usr/sbin/nologin
ircd:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:102:105:nonexistent:/usr/sbin/nologin
systemd-timesync:x:103:106:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
syslog:x:104:111:/home/syslog:/usr/sbin/nologin
apt:x:105:65534:/nonexistent:/usr/sbin/nologin
tpm:x:106:113:TPM software stack,,,:/var/lib/tpm:/bin/false
uidd:x:107:116:/run/uidd:/usr/sbin/nologin
systemd-userspace:x:108:117:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
```

- Using “Passwd” command to create or update password.

```
hariharan@hariharan-VirtualBox: /home$ passwd hariharan
Changing password for hariharan.
Current password:
New password:
Retype new password:
passwd: password updated successfully
hariharan@hariharan-VirtualBox: /home$
```

- Using “Cat/etc/group | grep” command to create a group of developer.


```
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep jarvis
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep jarvis
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep jarvis
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep users
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep hariharan
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep hari
hariharan@hariharan-VirtualBox: /home$ cat /etc/ os-release
cat: /etc/: Is a directory
cat: os-release: No such file or directory
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep users
hariharan@hariharan-VirtualBox: /home$
```

- Using “Cat/etc/os-release” command to os specs.

```
hariharan@hariharan-VirtualBox: /home$ cat /etc/ os-release
cat: /etc/: Is a directory
cat: os-release: No such file or directory
hariharan@hariharan-VirtualBox: /home$ cat /etc/group | grep users
hariharan@hariharan-VirtualBox: /home$ cat /etc/os-release
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
hariharan@hariharan-VirtualBox: /home$
```

- Using “chmod ug+rw file2” command and “ls -l” command to use read and write of an user and group of file 2.

The screenshot shows a terminal window titled "harhiaran@harhiaran-VirtualBox: ~/harhiaran". The user has executed several commands to create files and directories:

- `ls -l` shows existing files: `file1`, `file2`, `file3`, and `file4`.
- `mkdir harhiaran` creates a new directory.
- `mv file1 harhiaran` moves `file1` into the `harhiaran` directory.
- `mv file2 harhiaran` moves `file2` into the `harhiaran` directory.
- `mv file3 harhiaran` moves `file3` into the `harhiaran` directory.
- `mv file4 harhiaran` moves `file4` into the `harhiaran` directory.
- `cd harhiaran` changes the current directory to `harhiaran`.
- `ls` shows the files now inside the `harhiaran` directory: `file1`, `file2`, `file3`, and `file4`.
- `ls -l` shows the permissions of the files in the `harhiaran` directory: `-rw-rw-r-- 1 harhiaran harhiaran 0 Dec 27 13:56 file1`, `-rw-rw-r-- 1 harhiaran harhiaran 0 Dec 27 13:56 file2`, `-rw-rw-r-- 1 harhiaran harhiaran 0 Dec 27 13:56 file3`, and `-rw-rw-r-- 1 harhiaran harhiaran 0 Dec 27 13:56 file4`.
- `chmod ug+rw file2` changes the permissions of `file2` to `-rw-rw-r--`.
- `ls` shows the files in the `harhiaran` directory: `file1`, `file2`, `file3`, and `file4`.

- Using “chmod u-r file2” and “ls -l” command to permission denied of user, groups and others.

The screenshot displays a Kali Linux desktop environment. At the top, the menu bar includes 'Sunday (Planning)', 'Oracle VM VirtualBox', and a standard application menu with 'File', 'Machine', 'View', 'Input', 'Devices', and 'Help'. The desktop background is a dark purple gradient. A terminal window is open in the center, showing the following commands and output:

```

hariharan@hariharan-VirtualBox: ~$ ls
.  ..  Desktop  Downloads  file3  file.txt  hariharan  Pictures  snap  Videos
mnts  file2      file4  file.txt.save  Music  Public  Templates

hariharan@hariharan-VirtualBox: ~$ mv file2 hariharan

hariharan@hariharan-VirtualBox: ~$ ls
.  ..  Desktop  Downloads  file3  file4  file.txt  file.txt.save  hariharan  Music  Pictures  Public  snap  Templates  Videos

hariharan@hariharan-VirtualBox: ~$ mv file3 file4 hariharan

hariharan@hariharan-VirtualBox: ~$ ls
.  ..  Desktop  Downloads  file.txt  file.txt.save  hariharan  Music  Pictures  Public  snap  Templates  Videos

hariharan@hariharan-VirtualBox: ~$ cd hariharan

hariharan@hariharan-VirtualBox: ~/hariharan$ ls
file2  file3  file4

hariharan@hariharan-VirtualBox: ~/hariharan$ ls -l
total 0
-rwxr-xr-x 1 hariharan hariharan 0 Dec 27 13:56 file1
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file2
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file3
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file4

hariharan@hariharan-VirtualBox: ~/hariharan$ chmod ug+rw file2

hariharan@hariharan-VirtualBox: ~/hariharan$ ls
file2  file3  file4

hariharan@hariharan-VirtualBox: ~/hariharan$ chmod u-r file2

hariharan@hariharan-VirtualBox: ~/hariharan$ ls
file1  file2  file3  file4

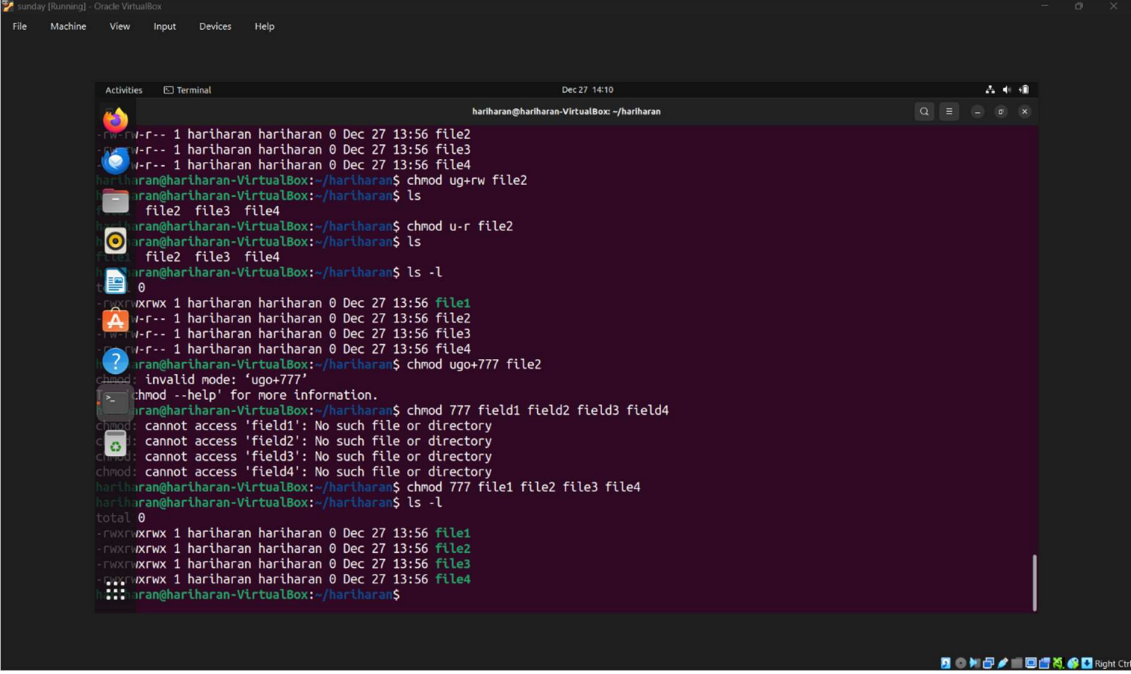
hariharan@hariharan-VirtualBox: ~/hariharan$ ls -l
total 0
-rwxr-xrwx 1 hariharan hariharan 0 Dec 27 13:56 file1
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file2
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file3
-rw-r--r-- 1 hariharan hariharan 0 Dec 27 13:56 file4

hariharan@hariharan-VirtualBox: ~/hariharan$

```

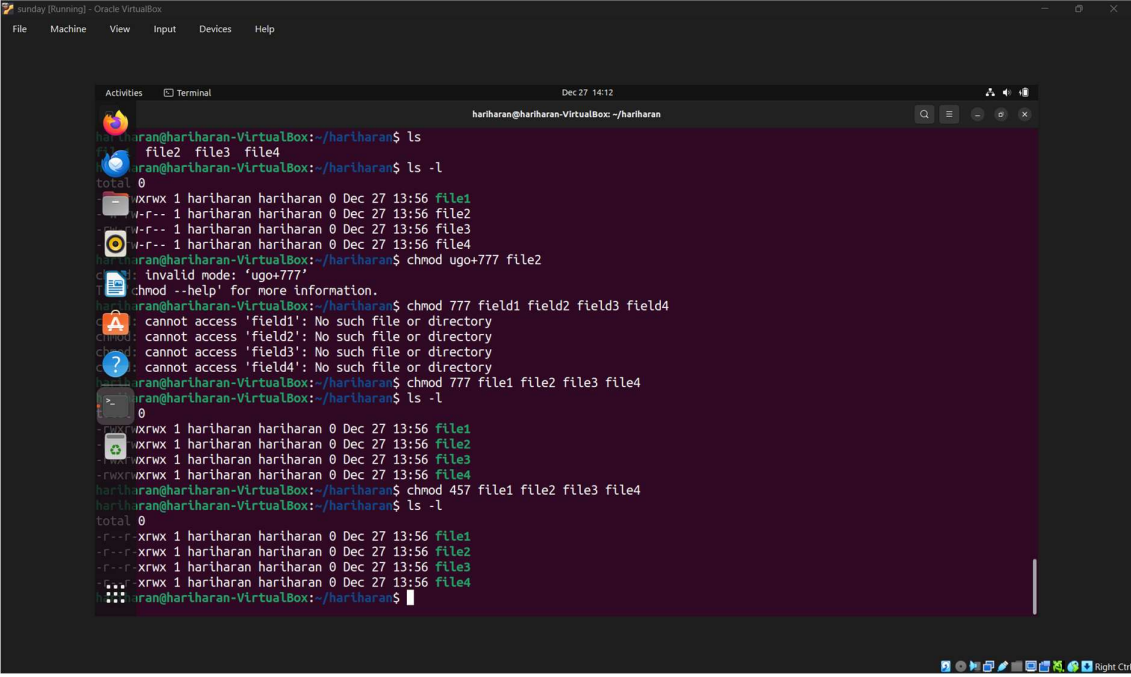
A file explorer window is also open, showing the contents of the ~/hariharan directory. It lists files file1, file2, file3, and file4 with their respective permissions and sizes. The file explorer window has a search bar and a sidebar with navigation options like Home, Recent, and Favorites.

- Using “`chmod 777 file1 file2 file3 file4`” command and “`ls -l`” command to read, write and execute of an user, group and others.



```
hartharan@hartharan-VirtualBox: ~/hartharan
hartharan@hartharan-VirtualBox:~/hartharan$ ls
file2  file3  file4
hartharan@hartharan-VirtualBox:~/hartharan$ chmod ugo+rw file2
hartharan@hartharan-VirtualBox:~/hartharan$ ls
file2  file3  file4
hartharan@hartharan-VirtualBox:~/hartharan$ chmod u+r file2
hartharan@hartharan-VirtualBox:~/hartharan$ ls
file2  file3  file4
hartharan@hartharan-VirtualBox:~/hartharan$ ls -l
total 0
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file1
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file2
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file3
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file4
hartharan@hartharan-VirtualBox:~/hartharan$ chmod ugo+777 file2
chmod: invalid mode: 'ugo+777'
chmod --help for more information.
hartharan@hartharan-VirtualBox:~/hartharan$ chmod 777 field1 field2 field3 field4
chmod: cannot access 'field1': No such file or directory
chmod: cannot access 'field2': No such file or directory
chmod: cannot access 'field3': No such file or directory
chmod: cannot access 'field4': No such file or directory
hartharan@hartharan-VirtualBox:~/hartharan$ chmod 777 file1 file2 file3 file4
hartharan@hartharan-VirtualBox:~/hartharan$ ls -l
total 0
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file1
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file2
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file3
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file4
hartharan@hartharan-VirtualBox:~/hartharan$
```

- Using “chmod 457 file1 file2 file3 file4” command and “ls -l” command to read,write and execute of an user,group and others.



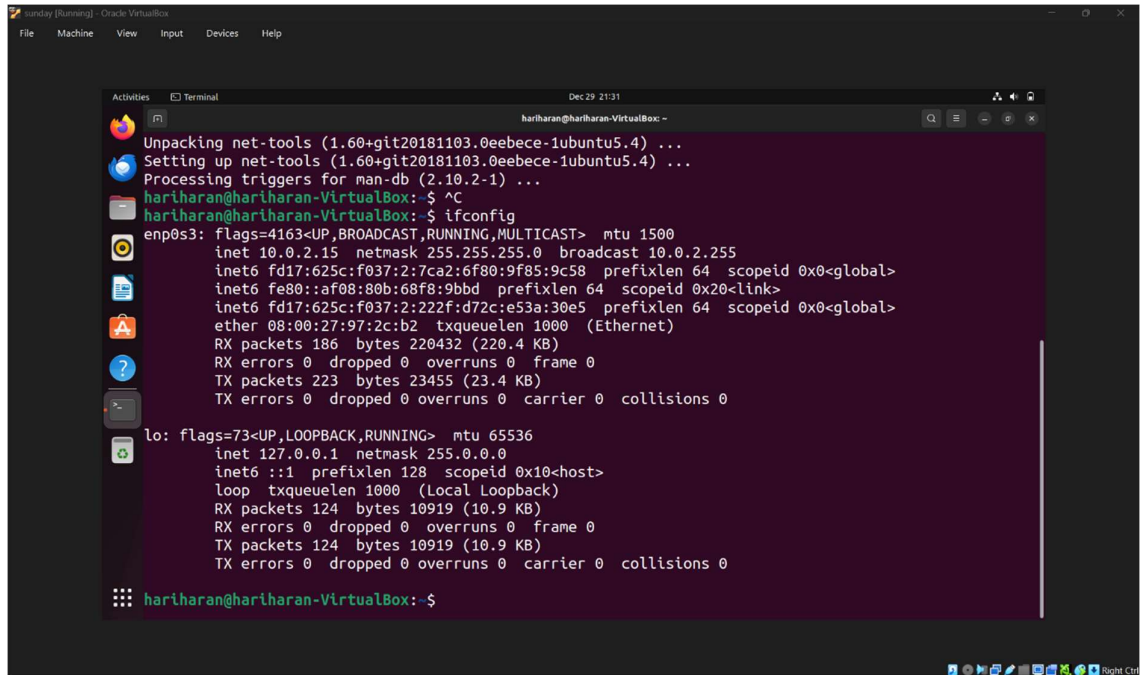
```
hartharan@hartharan-VirtualBox: ~/hartharan
hartharan@hartharan-VirtualBox:~/hartharan$ ls
file2  file3  file4
hartharan@hartharan-VirtualBox:~/hartharan$ ls -l
total 0
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file1
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file2
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file3
-rw-r--r-- 1 hartharan hartharan 0 Dec 27 13:56 file4
hartharan@hartharan-VirtualBox:~/hartharan$ chmod ugo+777 file2
chmod: invalid mode: 'ugo+777'
chmod --help for more information.
hartharan@hartharan-VirtualBox:~/hartharan$ chmod 777 field1 field2 field3 field4
chmod: cannot access 'field1': No such file or directory
chmod: cannot access 'field2': No such file or directory
chmod: cannot access 'field3': No such file or directory
chmod: cannot access 'field4': No such file or directory
hartharan@hartharan-VirtualBox:~/hartharan$ chmod 777 file1 file2 file3 file4
hartharan@hartharan-VirtualBox:~/hartharan$ ls -l
total 0
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file1
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file2
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file3
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file4
hartharan@hartharan-VirtualBox:~/hartharan$ chmod 457 file1 file2 file3 file4
hartharan@hartharan-VirtualBox:~/hartharan$ ls -l
total 0
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file1
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file2
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file3
-rwxr-xrwx 1 hartharan hartharan 0 Dec 27 13:56 file4
hartharan@hartharan-VirtualBox:~/hartharan$
```

- Using “chmod 000 file1 file2 file3 file4” command and “ls -l” command to empty of an user,group and others.

```
harisharan@harisharan-VirtualBox: ~/harisharan
$ chmod ugo+777 file2
chmod: invalid mode: 'ugo+777'
Try 'chmod --help' for more information.
harisharan@harisharan-VirtualBox:~/harisharan$ chmod 777 field1 field2 field3 field4
chmod: cannot access 'field1': No such file or directory
chmod: cannot access 'field2': No such file or directory
chmod: cannot access 'field3': No such file or directory
chmod: cannot access 'field4': No such file or directory
harisharan@harisharan-VirtualBox:~/harisharan$ chmod 777 file1 file2 file3 file4
harisharan@harisharan-VirtualBox:~/harisharan$ ls -l
total 0
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file1
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file2
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file3
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file4
harisharan@harisharan-VirtualBox:~/harisharan$ chmod 457 file1 file2 file3 file4
harisharan@harisharan-VirtualBox:~/harisharan$ ls -l
total 0
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file1
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file2
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file3
-rwxrwx 1 harisharan harisharan 0 Dec 27 13:56 file4
harisharan@harisharan-VirtualBox:~/harisharan$ chmod 000 file1 file2 file3 file4
harisharan@harisharan-VirtualBox:~/harisharan$ ls -l
total 0
----- 1 harisharan harisharan 0 Dec 27 13:56 file1
----- 1 harisharan harisharan 0 Dec 27 13:56 file2
----- 1 harisharan harisharan 0 Dec 27 13:56 file3
----- 1 harisharan harisharan 0 Dec 27 13:56 file4
harisharan@harisharan-VirtualBox:~/harisharan$
```

NETWORKING COMMANDS, DISK MANAGEMENT COMMANDS , FILE HIERARCHY STRUCTURE , ADD THE USER IN GROUP USING THE USERMOD

- Using “ifconfig” command use to identify the ip address.



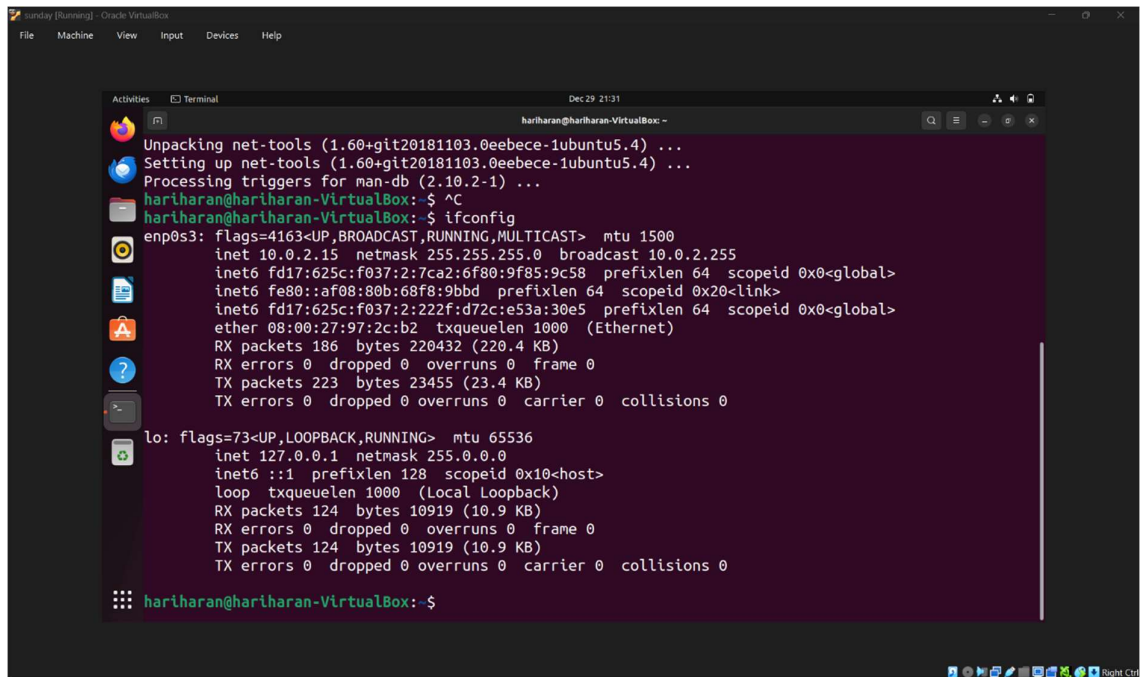
The screenshot shows a terminal window titled "hariharan@hariharan-VirtualBox" with the following output:

```
Unpacking net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Setting up net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Processing triggers for man-db (2.10.2-1) ...
hariharan@hariharan-VirtualBox:~$ ^C
hariharan@hariharan-VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:7ca2:6f80:9f85:9c58 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::af08:80b:68f8:9bbd prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:222f:d72c:e53a:30e5 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:97:2c:b2 txqueuelen 1000 (Ethernet)
    RX packets 186 bytes 220432 (220.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 223 bytes 23455 (23.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 124 bytes 10919 (10.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 124 bytes 10919 (10.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

hariharan@hariharan-VirtualBox:~$
```

- Using “ping” command use to testing network connectivity.

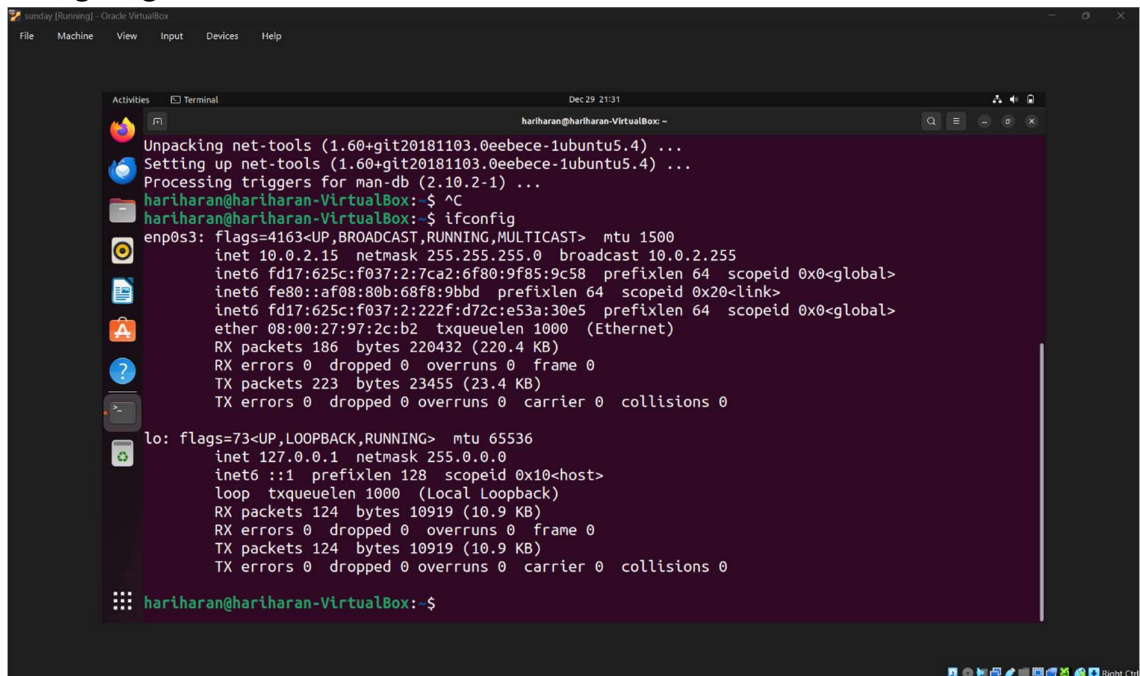


```
Unpacking net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Setting up net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Processing triggers for man-db (2.10.2-1) ...
hariharan@hariharan-VirtualBox: ~$ ^C
hariharan@hariharan-VirtualBox: ~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:7ca2:6f80:9f85:9c58 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::af08:80b:68f8:9bbd prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:222f:d72c:e53a:30e5 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:97:2c:b2 txqueuelen 1000 (Ethernet)
    RX packets 186 bytes 220432 (220.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 223 bytes 23455 (23.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 124 bytes 10919 (10.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 124 bytes 10919 (10.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

hariharan@hariharan-VirtualBox: ~$
```

- Using “wget” command to use to test website link of network download.

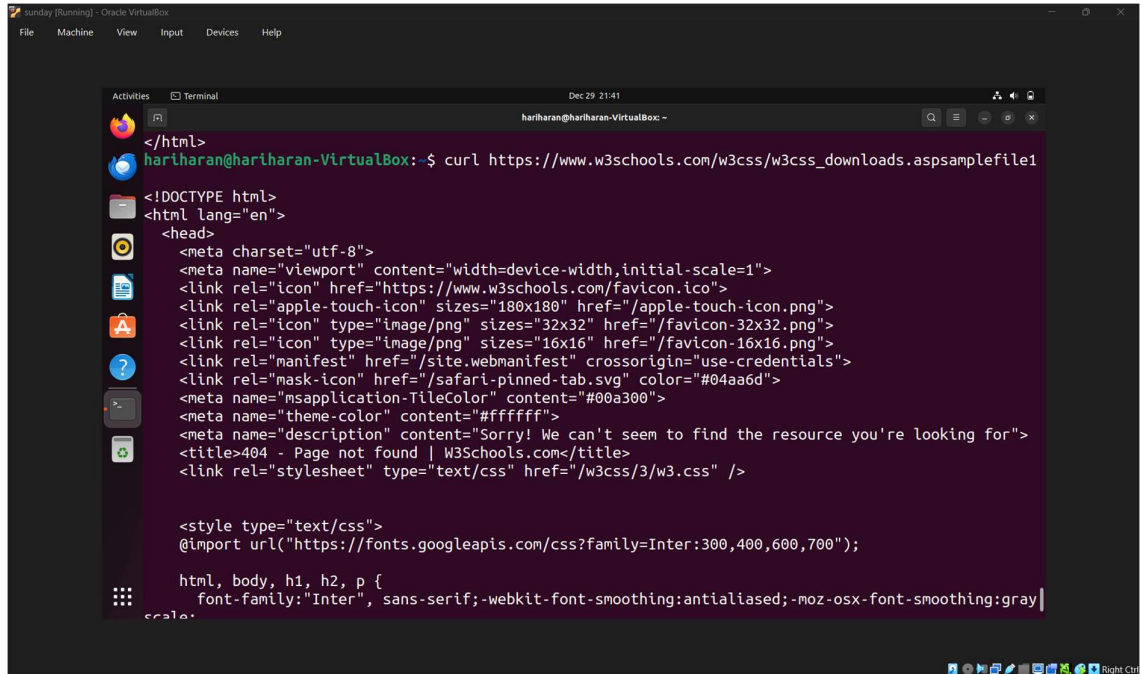


```
Unpacking net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Setting up net-tools (1.60+git20181103.0eebece-1ubuntu5.4) ...
Processing triggers for man-db (2.10.2-1) ...
hariharan@hariharan-VirtualBox: ~$ ^C
hariharan@hariharan-VirtualBox: ~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fd17:625c:f037:2:7ca2:6f80:9f85:9c58 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::af08:80b:68f8:9bbd prefixlen 64 scopeid 0x20<link>
    inet6 fd17:625c:f037:2:222f:d72c:e53a:30e5 prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:97:2c:b2 txqueuelen 1000 (Ethernet)
    RX packets 186 bytes 220432 (220.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 223 bytes 23455 (23.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 124 bytes 10919 (10.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 124 bytes 10919 (10.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

hariharan@hariharan-VirtualBox: ~$
```

- Using “curl” command to use to package of network download.



```
sunday [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Activities Terminal Dec 29 21:41
hariharan@hariharan-VirtualBox: ~

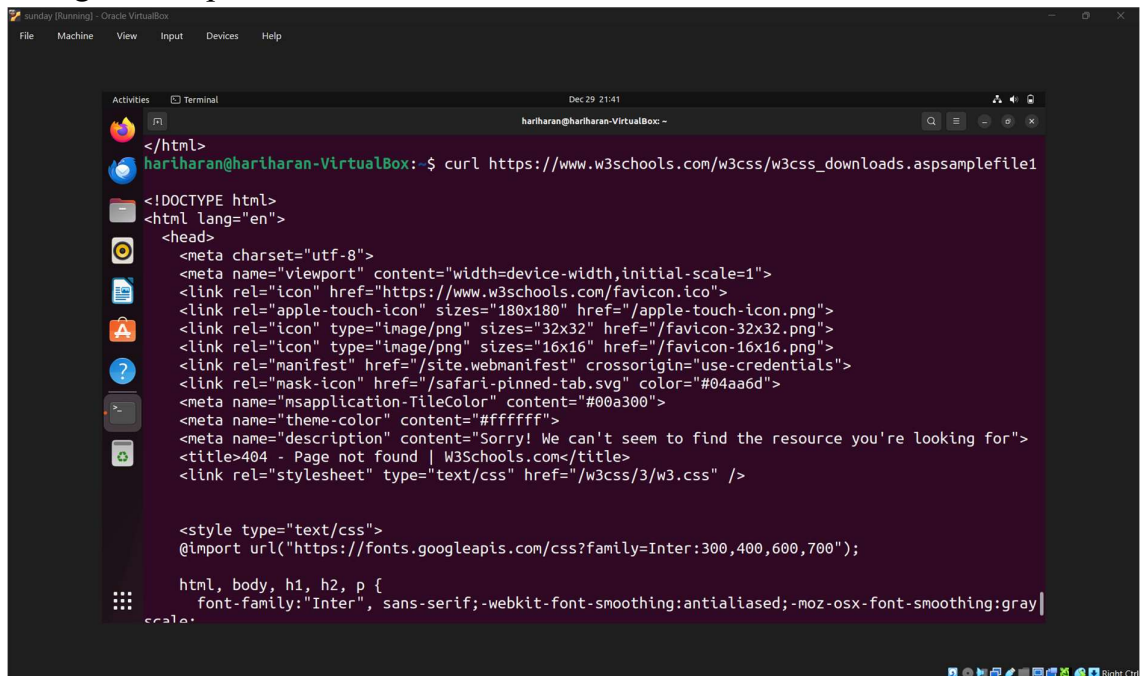
hariharan@hariharan-VirtualBox: ~$ curl https://www.w3schools.com/w3css/w3css_downloads.aspsamplefile1

</html>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<link rel="icon" href="https://www.w3schools.com/favicon.ico">
<link rel="apple-touch-icon" sizes="180x180" href="/apple-touch-icon.png">
<link rel="icon" type="image/png" sizes="32x32" href="/favicon-32x32.png">
<link rel="icon" type="image/png" sizes="16x16" href="/favicon-16x16.png">
<link rel="manifest" href="/site.webmanifest" crossorigin="use-credentials">
<link rel="mask-icon" href="/safari-pinned-tab.svg" color="#04aa6d">
<meta name="msapplication-TileColor" content="#00a300">
<meta name="theme-color" content="#ffffff">
<meta name="description" content="Sorry! We can't seem to find the resource you're looking for">
<title>404 - Page not found | W3Schools.com</title>
<link rel="stylesheet" type="text/css" href="/w3css/3/w3.css" />

<style type="text/css">
@import url("https://fonts.googleapis.com/css?family=Inter:300,400,600,700");

html, body, h1, h2, p {
font-family:"Inter", sans-serif;-webkit-font-smoothing:antialiased;-moz-osx-font-smoothing:gray
scale:
```

- Using “sudo apt install telnet” to install an telnet.



```
sunday [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Activities Terminal Dec 29 21:41
hariharan@hariharan-VirtualBox: ~

hariharan@hariharan-VirtualBox: ~$ curl https://www.w3schools.com/w3css/w3css_downloads.aspsamplefile1

</html>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<link rel="icon" href="https://www.w3schools.com/favicon.ico">
<link rel="apple-touch-icon" sizes="180x180" href="/apple-touch-icon.png">
<link rel="icon" type="image/png" sizes="32x32" href="/favicon-32x32.png">
<link rel="icon" type="image/png" sizes="16x16" href="/favicon-16x16.png">
<link rel="manifest" href="/site.webmanifest" crossorigin="use-credentials">
<link rel="mask-icon" href="/safari-pinned-tab.svg" color="#04aa6d">
<meta name="msapplication-TileColor" content="#00a300">
<meta name="theme-color" content="#ffffff">
<meta name="description" content="Sorry! We can't seem to find the resource you're looking for">
<title>404 - Page not found | W3Schools.com</title>
<link rel="stylesheet" type="text/css" href="/w3css/3/w3.css" />

<style type="text/css">
@import url("https://fonts.googleapis.com/css?family=Inter:300,400,600,700");

html, body, h1, h2, p {
font-family:"Inter", sans-serif;-webkit-font-smoothing:antialiased;-moz-osx-font-smoothing:gray
scale:
```

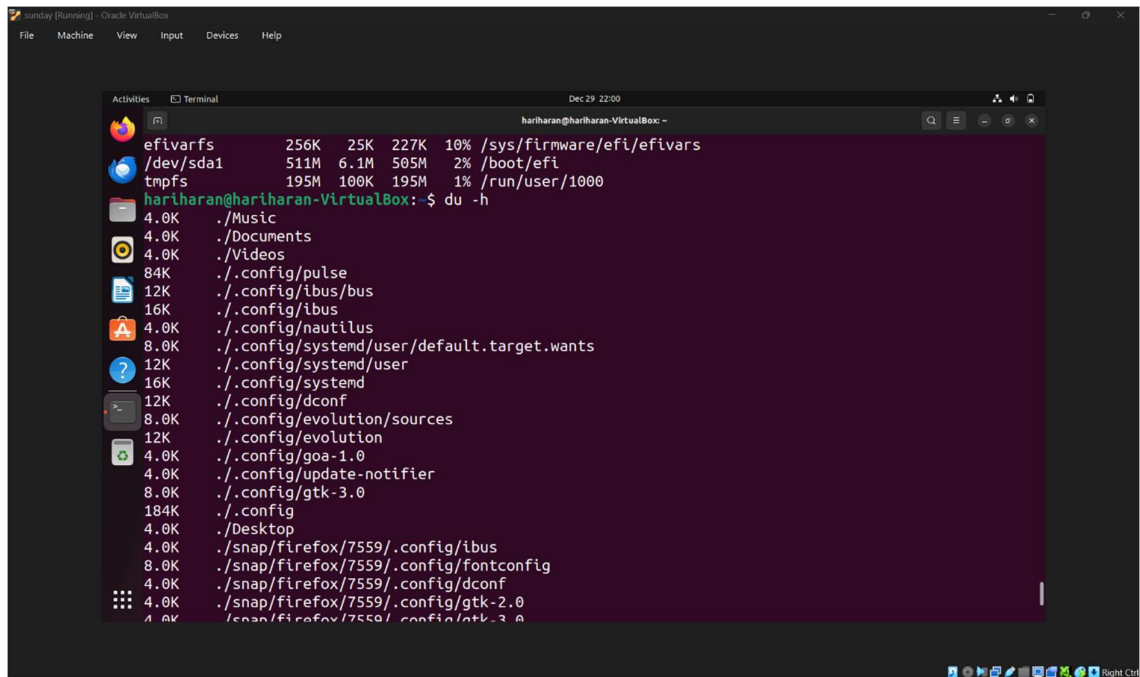
- Using “lsblk” command to list block statement.

```
Building dependency tree... Done
Reading state information... Done
telnet is already the newest version (0.17-44build1).
telnet set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 306 not upgraded.
hariharan@hariharan-VirtualBox:~$ telnet 49.206.10.74 22
Trying 49.206.10.74...
telnet: Unable to connect to remote host: Connection refused
hariharan@hariharan-VirtualBox:~$ lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0     4K  1 loop /snap/bare/5
loop1       7:1    0   74.3M  1 loop /snap/core22/1612
loop2       7:2    0  271.2M  1 loop /snap/firefox/4848
loop3       7:3    0  505.1M  1 loop /snap/gnome-42-2204/176
loop4       7:4    0  250.8M  1 loop /snap/firefox/7559
loop5       7:5    0   91.7M  1 loop /snap/gtk-common-themes/1535
loop6       7:6    0   12.9M  1 loop /snap/snap-store/1113
loop7       7:7    0   38.8M  1 loop /snap/snapd/21759
loop8       7:8    0   50.9M  1 loop /snap/snapd/25577
loop9       7:9    0   500K  1 loop /snap/snapd-desktop-integration/178
loop10      7:10   0   576K  1 loop /snap/snapd-desktop-integration/315
sda         8:0    0    25G  0 disk 
├─sda1      8:1    0   512M  0 part /boot/efi
└─sda2      8:2    0   24.5G  0 part /
sr0        11:0    1  1024M  0 rom
```

- Using “df -h” command to use disk free humanreadable.

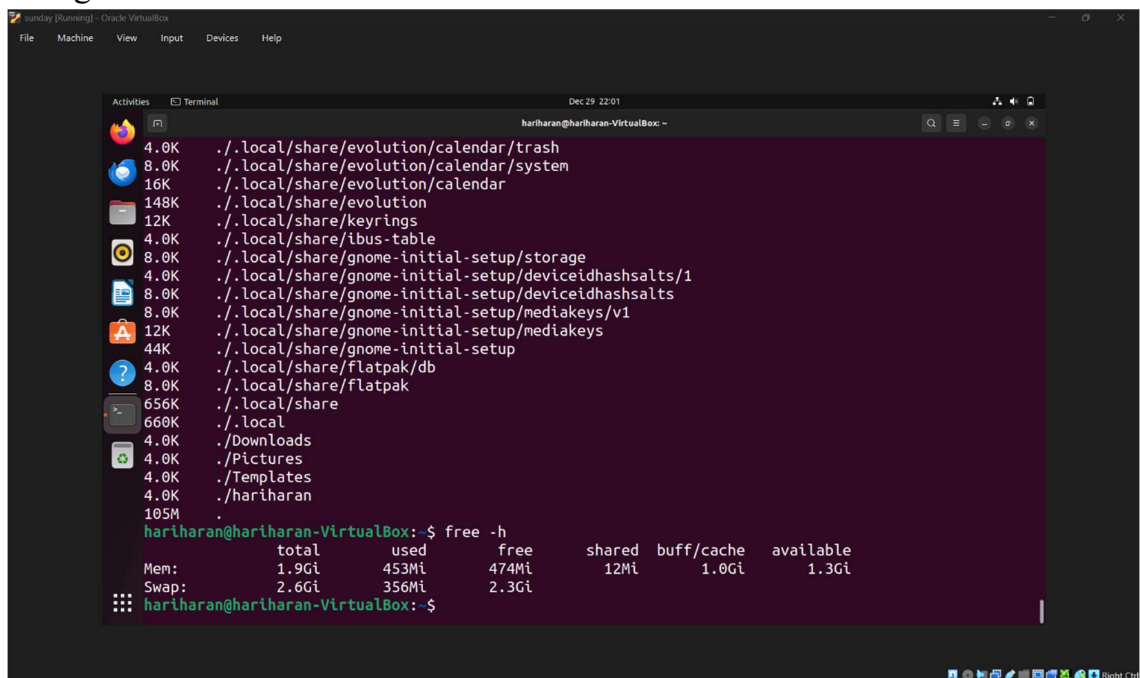
```
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0       7:0    0     4K  1 loop /snap/bare/5
loop1       7:1    0   74.3M  1 loop /snap/core22/1612
loop2       7:2    0  271.2M  1 loop /snap/firefox/4848
loop3       7:3    0  505.1M  1 loop /snap/gnome-42-2204/176
loop4       7:4    0  250.8M  1 loop /snap/firefox/7559
loop5       7:5    0   91.7M  1 loop /snap/gtk-common-themes/1535
loop6       7:6    0   12.9M  1 loop /snap/snap-store/1113
loop7       7:7    0   38.8M  1 loop /snap/snapd/21759
loop8       7:8    0   50.9M  1 loop /snap/snapd/25577
loop9       7:9    0   500K  1 loop /snap/snapd-desktop-integration/178
loop10      7:10   0   576K  1 loop /snap/snapd-desktop-integration/315
sda         8:0    0    25G  0 disk 
├─sda1      8:1    0   512M  0 part /boot/efi
└─sda2      8:2    0   24.5G  0 part /
sr0        11:0    1  1024M  0 rom
hariharan@hariharan-VirtualBox:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           195M  1.5M  194M   1% /run
/dev/sda2       24G   13G   11G  56% /
tmpfs           975M    0  975M   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
efivarfs       256K  25K  227K  10% /sys/firmware/efi/efivars
/dev/sda1       511M  6.1M  505M   2% /boot/efi
tmpfs           195M  100K  195M   1% /run/user/1000
```

- Using “du -h” command to use disk usage humanreadable.



```
efivarfs      256K   25K   227K   10% /sys/firmware/efi/efivars
/dev/sda1     511M   6.1M   505M    2% /boot/efi
tmpfs         195M   100K   195M    1% /run/user/1000
hariharan@hariharan-VirtualBox:~$ du -h
4.0K      ./Music
4.0K      ./Documents
4.0K      ./Videos
84K       ./config/pulse
12K       ./config/ibus/bus
16K       ./config/ibus
4.0K      ./config/nautilus
8.0K      ./config/systemd/user/default.target.wants
12K       ./config/systemd/user
16K       ./config/systemd
12K       ./config/dconf
8.0K      ./config/evolution/sources
12K       ./config/evolution
4.0K      ./config/goa-1.0
4.0K      ./config/update-notifier
8.0K      ./config/gtk-3.0
184K      ./config
4.0K      ./Desktop
4.0K      ./snap/firefox/7559/.config/ibus
8.0K      ./snap/firefox/7559/.config/fontconfig
4.0K      ./snap/firefox/7559/.config/dconf
4.0K      ./snap/firefox/7559/.config/gtk-2.0
4.0K      ./snap/firefox/7559/.config/gtk-3.0
```

- Using “free -h” command to use free humanreadable.



```
4.0K      ./local/share/evolution/calendar/trash
8.0K      ./local/share/evolution/calendar/system
16K       ./local/share/evolution/calendar
148K      ./local/share/evolution
12K       ./local/share/keyrings
4.0K      ./local/share/ibus-table
8.0K      ./local/share/gnome-initial-setup/storage
4.0K      ./local/share/gnome-initial-setup/deviceidhashsalts/1
8.0K      ./local/share/gnome-initial-setup/deviceidhashsalts
8.0K      ./local/share/gnome-initial-setup/mediakeys/v1
12K       ./local/share/gnome-initial-setup/mediakeys
44K       ./local/share/gnome-initial-setup
4.0K      ./local/share/flatpak/db
8.0K      ./local/share/flatpak
656K      ./local/share
660K      ./local
4.0K      ./Downloads
4.0K      ./Pictures
4.0K      ./Templates
4.0K      ./hariharan
105M      .
hariharan@hariharan-VirtualBox:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          1.9Gi         453Mi        474Mi        12Mi        1.0Gi        1.3Gi
Swap:         2.6Gi         356Mi        2.3Gi
```

- Using “ps -ef | grep python” command to use process of an python

```
hariharan@hariharan-VirtualBox: ~  
$ du -sh /usr/share/gnome-initial-setup/storage  
8.0K /usr/share/gnome-initial-setup/storage  
$ du -sh /usr/share/gnome-initial-setup/deviceidhashsalts/1  
4.0K /usr/share/gnome-initial-setup/deviceidhashsalts/1  
$ du -sh /usr/share/gnome-initial-setup/deviceidhashsalts  
8.0K /usr/share/gnome-initial-setup/deviceidhashsalts  
$ du -sh /usr/share/gnome-initial-setup/mediakeys/v1  
8.0K /usr/share/gnome-initial-setup/mediakeys/v1  
$ du -sh /usr/share/gnome-initial-setup/mediakeys  
12K /usr/share/gnome-initial-setup/mediakeys  
$ du -sh /usr/share/gnome-initial-setup  
44K /usr/share/gnome-initial-setup  
$ du -sh /usr/share/flatpak/db  
4.0K /usr/share/flatpak/db  
$ du -sh /usr/share/flatpak  
8.0K /usr/share/flatpak  
$ du -sh /usr/share  
656K /usr/share  
$ du -sh /usr/local  
660K /usr/local  
$ du -sh /usr/downloads  
4.0K /usr/downloads  
$ du -sh /usr/pictures  
4.0K /usr/pictures  
$ du -sh /usr/templates  
4.0K /usr/templates  
$ du -sh /usr/hariharan  
4.0K /usr/hariharan  
$ du -sh /usr  
105M /usr  
$ free -h  
             total        used        free      shared  buff/cache   available  
Mem:          1.9Gi         453Mi        474Mi        12Mi        1.0Gi        1.3Gi  
Swap:          2.6Gi         356Mi        2.3Gi  
$ ps -ef | grep python  
root          627      1  0 21:25 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --r  
un-startup-triggers  
root          731      1  0 21:25 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/u  
nattended-upgrade-shutdown --wait-for-signal  
harihar+    5125    1963  0 22:01 pts/0    00:00:00 grep --color=auto python  
$
```

- Using “uptime” command to use identify running time.

```
hariharan@hariharan-VirtualBox: ~  
$ free -h  
             total        used        free      shared  buff/cache   available  
Mem:          1.9Gi         453Mi        474Mi        12Mi        1.0Gi        1.3Gi  
Swap:          2.6Gi         356Mi        2.3Gi  
$ ps -ef | grep python  
root          627      1  0 21:25 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --r  
un-startup-triggers  
root          731      1  0 21:25 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/u  
nattended-upgrade-shutdown --wait-for-signal  
harihar+    5125    1963  0 22:01 pts/0    00:00:00 grep --color=auto python  
$ uptime  
22:03:42 up 38 min, 1 user,  load average: 0.13, 0.08, 0.17  
$
```

- Using “chown jarvis file3” command to use change ownership of file.

```
sunday [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Activities Terminal
hariharan@hariharan-VirtualBox: ~
Dec 29 22:26

drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Templates
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Videos
-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox:~$ chown jarvis file3
chown: changing ownership of 'file3': Operation not permitted
hariharan@hariharan-VirtualBox:~$ sudo chown jarvis file3
[sudo] password for hariharan:
chown: changing ownership of 'file3': Operation not permitted
hariharan@hariharan-VirtualBox:~$ ls -l
total 412
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Desktop
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Documents
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Downloads
-rw-rw-r-- 1 jarvis hariharan 0 Dec 29 19:45 file3
-rw-rw-r-- 1 hariharan hariharan 0 Dec 27 11:49 file.txt
-rw-rw-r-- 1 hariharan hariharan 2 Dec 27 11:56 file.txt.save
drwxrwxr-x 2 hariharan hariharan 4096 Dec 27 14:03 hariharan
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Music
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Pictures
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Public
drwx----- 4 hariharan hariharan 4096 Dec 29 21:37 snap
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Templates
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Videos
-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox:~$
```

- Using “chown :jarvis file3” command to use change ownership of group.

```
sunday [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

Activities Terminal
hariharan@hariharan-VirtualBox: ~
Dec 29 22:28

drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Videos
-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox:~$ chown: jarvis file3
Command 'chown:' not found, did you mean:
  command 'chown' from deb coreutils (8.32-4.1ubuntu1.2)
Try: sudo apt install <deb name>
hariharan@hariharan-VirtualBox:~$ chown :jarvis file3
chown: changing group of 'file3': Operation not permitted
hariharan@hariharan-VirtualBox:~$ sudo chown :jarvis file3
hariharan@hariharan-VirtualBox:~$ ls -l
total 412
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Desktop
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Documents
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Downloads
-rw-rw-r-- 1 jarvis jarvis 0 Dec 29 19:45 file3
-rw-rw-r-- 1 hariharan hariharan 0 Dec 27 11:49 file.txt
-rw-rw-r-- 1 hariharan hariharan 2 Dec 27 11:56 file.txt.save
drwxrwxr-x 2 hariharan hariharan 4096 Dec 27 14:03 hariharan
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Music
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Pictures
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Public
drwx----- 4 hariharan hariharan 4096 Dec 29 21:37 snap
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Templates
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Videos
-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox:~$
```

- Using “usermod -aG” command to user add to another group to get permission.

```
sunday [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

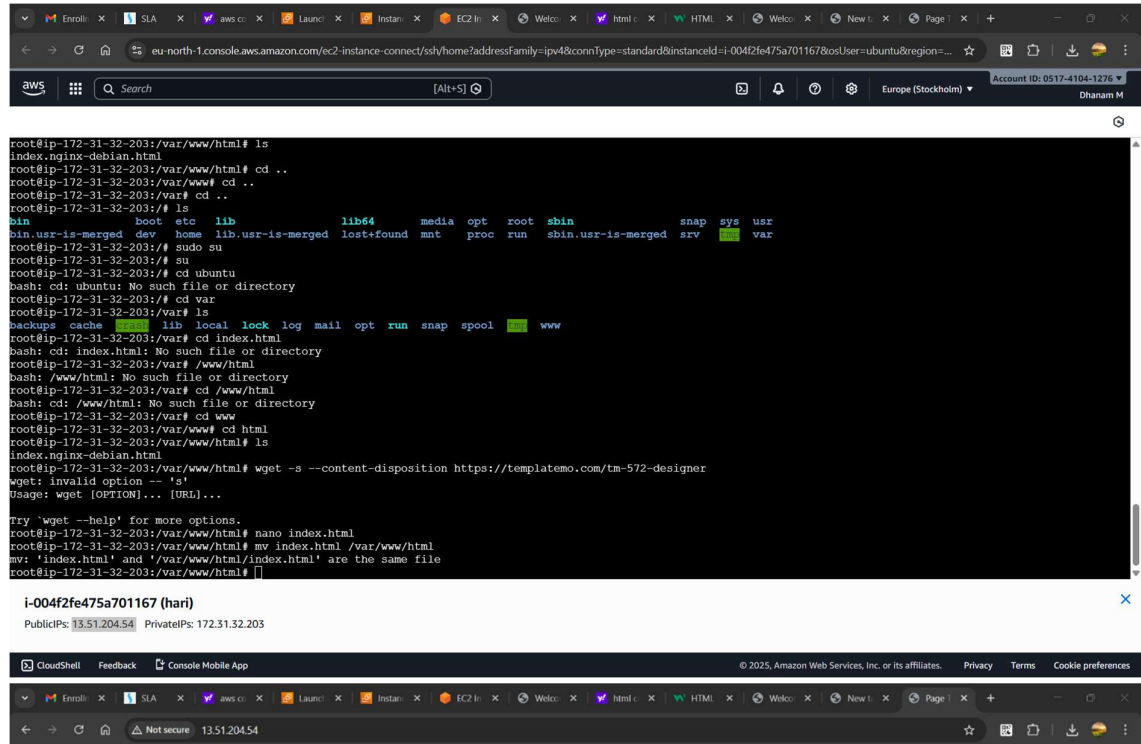
Activities Terminal Dec 29 22:32
hariharan@hariharan-VirtualBox: ~

-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox: $ cat /etc/group | grep jarvis
jarvis:x:1003:
hariharan@hariharan-VirtualBox: $ usermod -aG vijay jarvis
usermod: Permission denied.
usermod: cannot lock /etc/passwd; try again later.
hariharan@hariharan-VirtualBox: $ sudo usermod -aG vijay jarvis
hariharan@hariharan-VirtualBox: $ ls -l
total 412
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Desktop
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Documents
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Downloads
-rw-rw-r-- 1 jarvis jarvis 0 Dec 29 19:45 file3
-rw-rw-r-- 1 hariharan hariharan 0 Dec 27 11:49 file.txt
-rw-rw-r-- 1 hariharan hariharan 2 Dec 27 11:56 file.txt.save
drwxrwxr-x 2 hariharan hariharan 4096 Dec 27 14:03 hariharan
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Music
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Pictures
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Public
drwx----- 4 hariharan hariharan 4096 Dec 29 21:37 snap
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Templates
drwxr-xr-x 2 hariharan hariharan 4096 Dec 27 11:42 Videos
-rw-rw-r-- 1 hariharan hariharan 375078 Dec 29 21:39 w3css_downloads.asp
hariharan@hariharan-VirtualBox: $ groups jarvis
jarvis : jarvis vijay
hariharan@hariharan-VirtualBox: $
```

WEB HOSTING IN LINUX

WEB HOSTING IN UBUNTU:

- Using “nano index.html” command to edit the file to insert sample code of html “mv index.html /var/www/html” command to use move the file.



The screenshot displays the AWS CloudShell interface within a web browser. The terminal window shows a series of commands executed in an Ubuntu environment. The user starts by listing files in the /var/www/html directory, then attempts to navigate to /var/www/html and /var/www. After several failed attempts to navigate, the user successfully moves the index.html file to /var/www/html. The browser window shows the resulting web page, which displays "This is a Heading" and "This is a paragraph." The browser's address bar shows the URL "http://13.51.204.54".

```
root@ip-172-31-32-203:/var/www/html# ls
index.nginx-debian.html
root@ip-172-31-32-203:/var/www/html# cd ..
root@ip-172-31-32-203:/var/www# cd ..
root@ip-172-31-32-203:/var# cd ..
root@ip-172-31-32-203:/# ls
bin      boot     etc      lib      lib64    media   opt      root    /sbin    snap     sys      usr
bin.usr-is-merged  dev     home    lib.usr-is-merged  lost+found  mnt     proc    run     /sbin.usr-is-merged  srv      var
root@ip-172-31-32-203:/# su
root@ip-172-31-32-203:/# cd ubuntu
bash: cd: ubuntu: No such file or directory
root@ip-172-31-32-203:/# cd var
root@ip-172-31-32-203:/var# ls
backups  cache  lib  local  lock  log  mail  opt  run  snap  spool  www
root@ip-172-31-32-203:/var# cd index.html
bash: cd: index.html: No such file or directory
root@ip-172-31-32-203:/var# cd /var/www/html
bash: /var/www/html: No such file or directory
root@ip-172-31-32-203:/var# cd /var/www/html
bash: cd: /var/www/html: No such file or directory
root@ip-172-31-32-203:/var# cd www
root@ip-172-31-32-203:/var/www# cd html
root@ip-172-31-32-203:/var/www/html# ls
index.nginx-debian.html
root@ip-172-31-32-203:/var/www/html# wget -s --content-disposition https://templatemo.com/tm-572-designer
wget: invalid option -- 's'
Usage: wget [OPTION]... [URL]...

Try 'wget --help' for more options.
root@ip-172-31-32-203:/var/www/html# nano index.html
root@ip-172-31-32-203:/var/www/html# mv index.html /var/www/html
mv: 'index.html' and '/var/www/html/index.html' are the same file
root@ip-172-31-32-203:/var/www/html#
```

i-004f2fe475a701167 (hari)
PublicIPs: 13.51.204.54 PrivateIPs: 172.31.32.203

CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

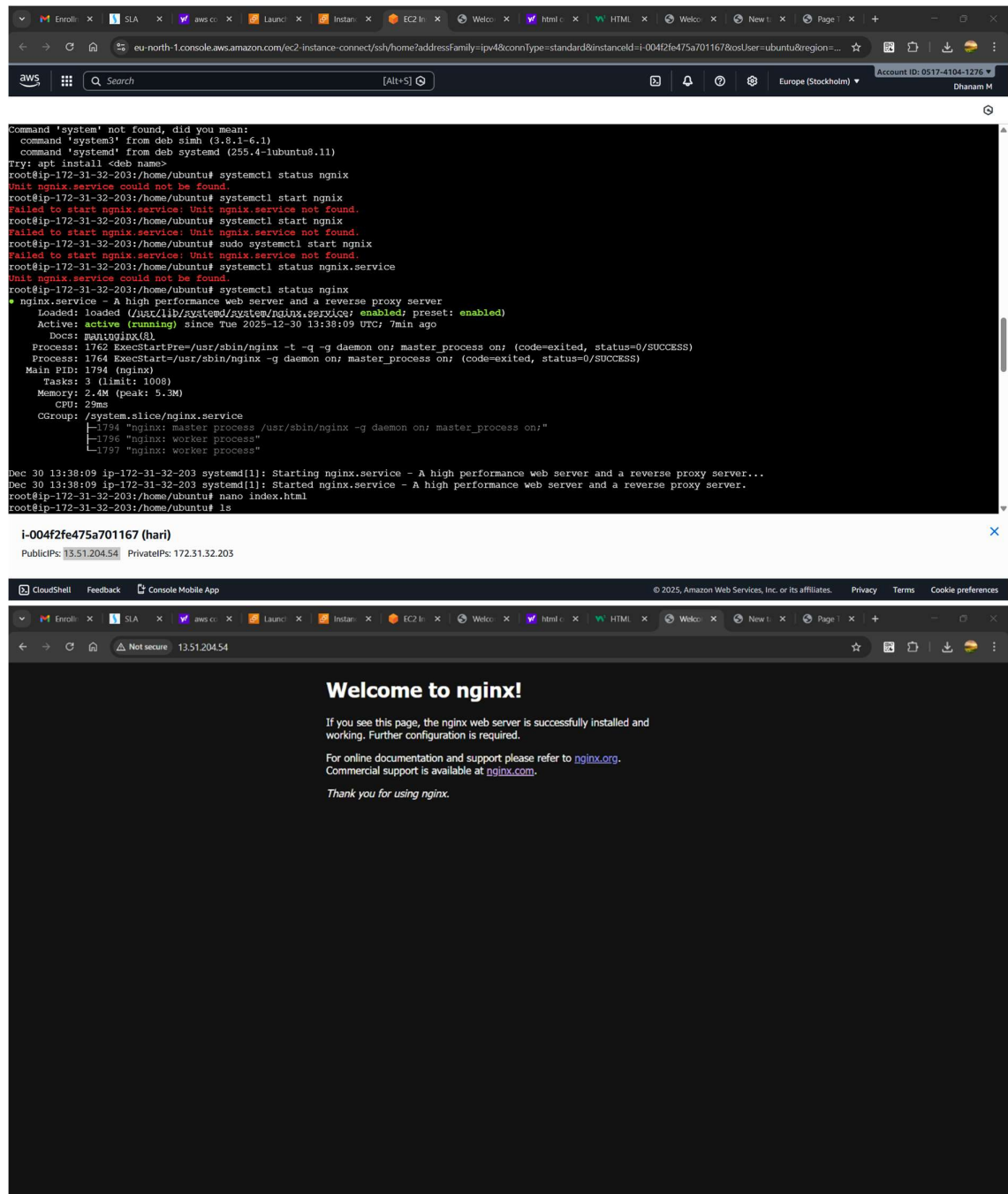
Enroll x SIA x aws c x Launch x Instan x EC2 In x Welco x html x HTML x Welco x New l x Page x +

Not secure 13.51.204.54

This is a Heading

This is a paragraph.

- Using “apt install nginx”command to install nginx “systemctl status nginx” command to check status is inactive “systemctl start nginx” command to activate it.



WEB HOSTING IN LINUX:

- Using “yum install httpd -y” command to install httpd and check with an public ip address.

The screenshot shows the AWS Management Console interface. At the top, there are tabs for 'Instance details | EC2 | eu-north-1' and 'EC2 Instance Connect | eu-north-1'. The browser address bar shows the URL: `eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=eu-north-1&connType=standard&instanceId=i-04d245bfd89664e4f&osUser=ec2-user&sshPort=...`. The AWS logo and search bar are visible on the left. The main content area displays a terminal window for an EC2 instance with ID `i-04d245bfd89664e4f` (name `hari`). The terminal shows the user `ec2-user` at `ip-172-31-37-2` running `sudo su` to become root. The root user then runs `yum update httpd -y`, which fails because the package is not installed. Next, `yum install httpd -y` is executed, which successfully installs `httpd` and its dependencies. A table lists the installed packages, their architectures, versions, repositories, and sizes. Below the table, a 'Transaction Summary' is provided. The instance's public and private IP addresses are also shown. At the bottom of the console, there are links for 'CloudShell', 'Feedback', and 'Console Mobile App', along with copyright information for Amazon Web Services.

```
[ec2-user@ip-172-31-37-2 ~]$ sudo su
[root@ip-172-31-37-2 ec2-user]# yum update httpd -y
Amazon Linux 2023 kernel Livepatch repository
Package httpd available, but not installed.
No match for argument: httpd
Error: No packages marked for upgrade.
[root@ip-172-31-37-2 ec2-user]# yum install httpd -y
Last metadata expiration check: 0:00:23 ago on Tue Dec 30 16:37:12 2025.
Dependencies resolved.
=====================================================================================================================================
Package                                     Architecture                               Version                                Repository                               Size
=====================================================================================================================================
Installing:
httpd                                       x86_64                                    2.4.65-1.amzn2023.0.2                 amazonlinux                              47 k
Installing dependencies:
apr                                         x86_64                                    1.7.5-1.amzn2023.0.4                 amazonlinux                              129 k
apr-util                                   x86_64                                    1.6.3-1.amzn2023.0.2                 amazonlinux                              97 k
apr-util-ldap                             x86_64                                    1.6.3-1.amzn2023.0.2                 amazonlinux                              13 k
generic-logos-httpd                       noarch                                    18.0.0-12.amzn2023.0.3              amazonlinux                              19 k
httpd-core                                 x86_64                                    2.4.65-1.amzn2023.0.2                 amazonlinux                              1.4 M
httpd-filesystem                          noarch                                    2.4.65-1.amzn2023.0.2                 amazonlinux                              13 k
httpd-tools                               x86_64                                    2.4.65-1.amzn2023.0.2                 amazonlinux                              81 k
libbrotli                                 x86_64                                    1.0.9-4.amzn2023.0.2                 amazonlinux                              315 k
mailcap                                    noarch                                    2.1.49-3.amzn2023.0.3              amazonlinux                              33 k
Installing weak dependencies:
apr-util-openssl                           x86_64                                    1.6.3-1.amzn2023.0.2                 amazonlinux                              15 k
mod_http2                                  x86_64                                    2.0.27-1.amzn2023.0.3              amazonlinux                              166 k
mod_lua                                    x86_64                                    2.4.65-1.amzn2023.0.2                 amazonlinux                              60 k
Transaction Summary
=====================================================================================================================================
```

i-04d245bfd89664e4f (hari)
PublicIPs: 16.171.31.5 PrivateIPs: 172.31.37.2

CloudShell Feedback Console Mobile App © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

13:203.205.1 Not secure 13.203.205.1 Incognito

It works!

- Using “systemctl status httpd” command to check status inactive
“systemctl start httpd” command to start activate it.

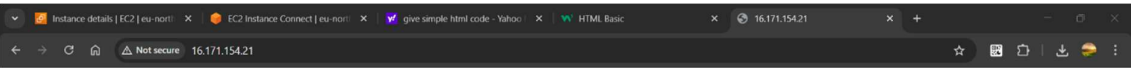
```
Complete!  
[root@ip-172-31-32-240 ec2-user]# systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: inactive (dead)  
     Docs: man:httpd.service(8)  
[root@ip-172-31-32-240 ec2-user]# systemctl start httpd  
[root@ip-172-31-32-240 ec2-user]# systemctl status httpd  
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: active (running) since Tue 2025-12-30 16:53:54 UTC; 22s ago  
     Docs: man:httpd.service(8)  
  Main PID: 25852 (httpd)  
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"  
    Tasks: 177 (limit: 1067)  
  Memory: 13.3M  
    CPU: 75ms  
   CGroup: /system.slice/httpd.service  
           └─25852 /usr/sbin/httpd -DFOREGROUND  
             └─25853 /usr/sbin/httpd -DFOREGROUND  
               └─25854 /usr/sbin/httpd -DFOREGROUND  
                 └─25855 /usr/sbin/httpd -DFOREGROUND  
                   └─25856 /usr/sbin/httpd -DFOREGROUND  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal httpd[25852]: Server configured, listening on: port 80  
[root@ip-172-31-32-240 ec2-user]#
```

i-067aa27ef9ce7adf6 (haran)
PublicIPs: 16.171.154.21 PrivateIPs: 172.31.32.240

- Using “nano index.html” command to edit the file “mv index.html /var/www/html/index.html” command to use move from it.

```
● httpd.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)  
   Active: active (running) since Tue 2025-12-30 16:53:54 UTC; 22s ago  
     Docs: man:httpd.service(8)  
  Main PID: 25852 (httpd)  
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"  
    Tasks: 177 (limit: 1067)  
  Memory: 13.3M  
    CPU: 75ms  
   CGroup: /system.slice/httpd.service  
           └─25852 /usr/sbin/httpd -DFOREGROUND  
             └─25853 /usr/sbin/httpd -DFOREGROUND  
               └─25854 /usr/sbin/httpd -DFOREGROUND  
                 └─25855 /usr/sbin/httpd -DFOREGROUND  
                   └─25856 /usr/sbin/httpd -DFOREGROUND  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.  
Dec 30 16:53:54 ip-172-31-32-240.eu-north-1.compute.internal httpd[25852]: Server configured, listening on: port 80  
[root@ip-172-31-32-240 ec2-user]# ls  
index.html  
[root@ip-172-31-32-240 ec2-user]# cat index.html  
<!DOCTYPE html>  
<html>  
<body>  
  
<h1>My First Heading</h1>  
<p>My first paragraph.</p>  
  
</body>  
</html>  
[root@ip-172-31-32-240 ec2-user]# mv index.html /var/www/html/index.html  
[root@ip-172-31-32-240 ec2-user]#
```

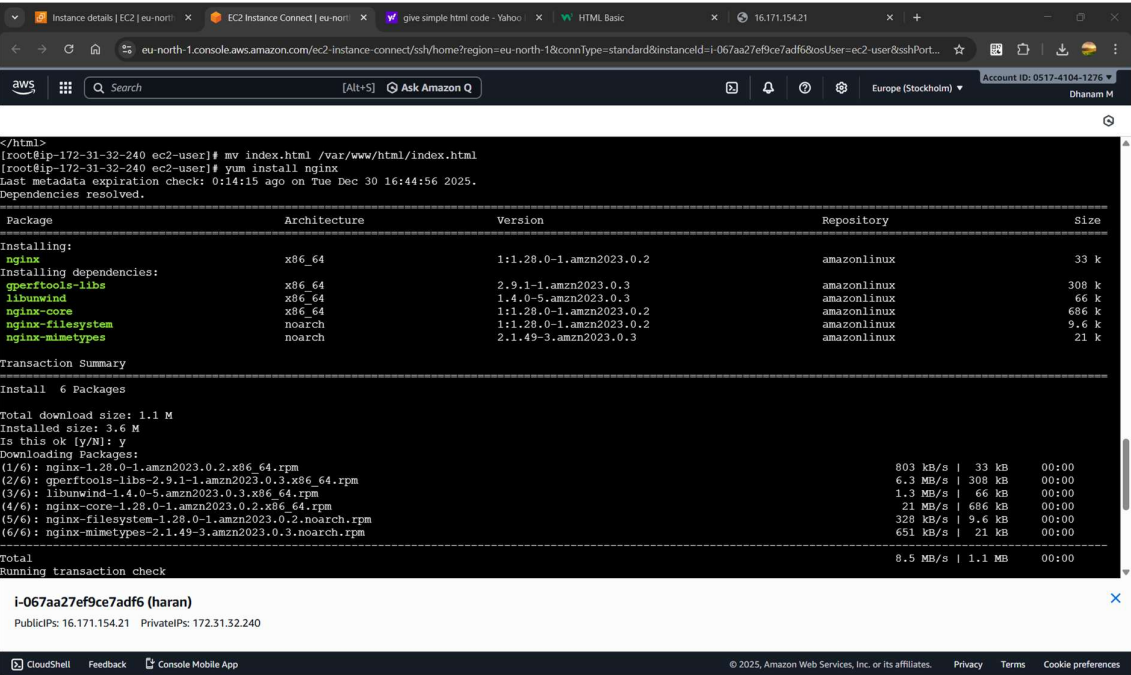
i-067aa27ef9ce7adf6 (haran)
PublicIPs: 16.171.154.21 PrivateIPs: 172.31.32.240

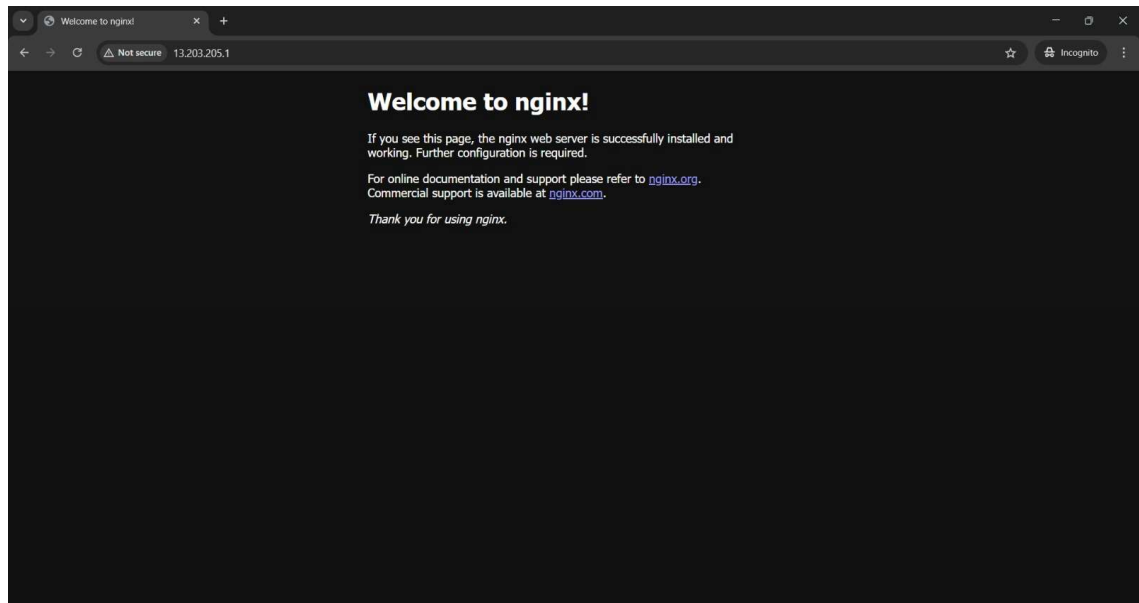


My First Heading

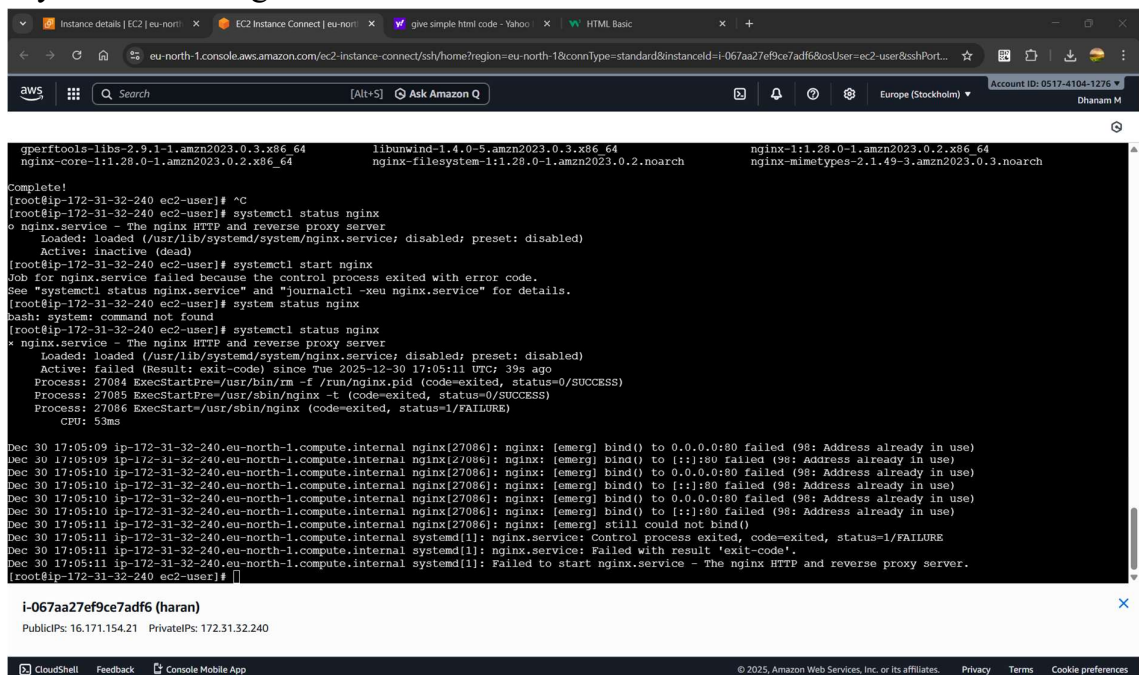
My first paragraph.

- Using “yum install nginx” command to install nginx and check with an public ip address.





- Using “systemctl status nginx” command to check status inactive
“systemctl start nginx” command to activate it.



- Using “nano index.html” command to edit the file “mv index.html /usr/share/nginx/html/index.html” command to use move it.

The screenshot shows a terminal window within the AWS Management Console. The user is logged in as 'ec2-user' on an EC2 instance. They run the following commands:

```
[root@ip-172-31-32-240 ec2-user]# ^C
[root@ip-172-31-32-240 ec2-user]# systemctl status nginx
o nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: inactive (dead)
[root@ip-172-31-32-240 ec2-user]# systemctl start nginx
Job for nginx.service failed because the control process exited with error code.
See "systemctl status nginx.service" and "journalctl -xeu nginx.service" for details.
[root@ip-172-31-32-240 ec2-user]# system status nginx
bash: system: command not found
[root@ip-172-31-32-240 ec2-user]# systemctl status nginx
* nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: failed (Result: exit-code) since Tue 2025-12-30 17:05:11 UTC; 39s ago
   Process: 27084 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
   Process: 27085 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
   Process: 27086 ExecStart=/usr/sbin/nginx (code=exited, status=1/FAILURE)
   CPU: 53ms

Dec 30 17:05:09 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Dec 30 17:05:10 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to :::80 failed (98: Address already in use)
Dec 30 17:05:10 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Dec 30 17:05:10 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to :::80 failed (98: Address already in use)
Dec 30 17:05:10 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to 0.0.0.0:80 failed (98: Address already in use)
Dec 30 17:05:10 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] bind() to :::80 failed (98: Address already in use)
Dec 30 17:05:11 ip-172-31-32-240.eu-north-1.compute.internal nginx[27086]: nginx: [emerg] still could not bind()
Dec 30 17:05:11 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: nginx.service: Control process exited, code=exited, status=1/FAILURE
Dec 30 17:05:11 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: nginx.service: Failed with result 'exit-code'.
Dec 30 17:05:11 ip-172-31-32-240.eu-north-1.compute.internal systemd[1]: Failed to start nginx.service - The nginx HTTP and reverse proxy server.

[root@ip-172-31-32-240 ec2-user]# nano index.html
[root@ip-172-31-32-240 ec2-user]# mv index.html /usr/share/nginx/html/index.html
mv: overwrite '/usr/share/nginx/html/index.html'? y
[root@ip-172-31-32-240 ec2-user]#
```

Below the terminal, the instance details are shown: **i-067aa27ef9ce7adf6 (haran)**, with Public IP **16.171.154.21** and Private IP **172.31.32.240**.

The bottom part of the screenshot shows a web browser window displaying the content of the index.html file, which contains the heading **My First Heading** and a paragraph **My first paragraph.**

- Using “`wget –s –content-disposition<copy link>`” command to install on zip file and using “`unzip`” use tap to unzip file.

```
--2025-12-30 17:15:38-- http://  
Resolving - (-)... failed: Name or service not known.  
wget: unable to resolve host address '-'  
--2025-12-30 17:15:38-- http://s/  
Resolving s (s)... failed: Name or service not known.  
wget: unable to resolve host address 's'  
--2025-12-30 17:15:38-- https://templatemo.com/download/templatemo_572_designer  
Resolving templatemo.com (templatemo.com)... 72.52.176.250  
Connecting to templatemo.com (templatemo.com)|72.52.176.250|:443... connected.  
HTTP request sent, awaiting response... 302 Moved Temporarily  
Location: https://templatemo.com/tm-zip-files-2020/templatemo_572_designer.zip [following]  
--2025-12-30 17:15:39-- https://templatemo.com/tm-zip-files-2020/templatemo_572_designer.zip  
Connecting to templatemo.com (templatemo.com)|72.52.176.250|:443... connected.  
HTTP request sent, awaiting response... 200 OK  
Length: 3549230 (3.4M) (application/zip)  
Saving to: 'templatemo_572_designer.zip'  
  
templatemo_572_designer.zip 100%[=====>] 3.38M 1.46MB/s in 2.3s  
  
2025-12-30 17:15:41 (1.46 MB/s) - 'templatemo_572_designer.zip' saved [3549230/3549230]  
  
FINISHED --2025-12-30 17:15:41--  
Total wall clock time: 3.7s  
Downloaded: 1 files, 3.4M in 2.3s (1.46 MB/s)  
[root@ip-172-31-32-240 ec2-user]# ls  
templatemo_572_designer.zip  
[root@ip-172-31-32-240 ec2-user]# unzip templatemo_572_designer.zip  
Archive: templatemo_572_designer.zip  
creating: templatemo_572_designer/  
inflating: templatemo_572_designer/about.html  
creating: templatemo_572_designer/assets/  
creating: templatemo_572_designer/assets/css/  
inflating: templatemo_572_designer/assets/css/animate.css  
inflating: templatemo_572_designer/assets/css/flex-slider.css
```

i-067aa27ef9ce7adf6 (haran)
PublicIPs: 16.171.154.21 PrivateIPs: 172.31.32.240

- Using “ls” command to list unzip file “mv * /var/www/html/” command to use move from it.

```
creating: templatemo_572_designer/assets/webfonts/  
inflating: templatemo_572_designer/assets/webfonts/fa-brands-400.ttf  
inflating: templatemo_572_designer/assets/webfonts/fa-brands-400.woff2  
inflating: templatemo_572_designer/assets/webfonts/fa-regular-400.ttf  
inflating: templatemo_572_designer/assets/webfonts/fa-regular-400.woff2  
inflating: templatemo_572_designer/assets/webfonts/fa-solid-900.ttf  
inflating: templatemo_572_designer/assets/webfonts/fa-solid-900.woff2  
inflating: templatemo_572_designer/assets/webfonts/fa-v4compatibility.ttf  
inflating: templatemo_572_designer/assets/webfonts/fa-v4compatibility.woff2  
inflating: templatemo_572_designer/contact.html  
inflating: templatemo_572_designer/explore.html  
inflating: templatemo_572_designer/index.html  
inflating: templatemo_572_designer/trending.html  
creating: templatemo_572_designer/vendor/  
creating: templatemo_572_designer/vendor/bootstrap/  
creating: templatemo_572_designer/vendor/bootstrap/css/  
inflating: templatemo_572_designer/vendor/bootstrap/css/bootstrap.min.css  
creating: templatemo_572_designer/vendor/bootstrap/js/  
inflating: templatemo_572_designer/vendor/bootstrap/js/bootstrap.min.js  
creating: templatemo_572_designer/vendor/jquery/  
inflating: templatemo_572_designer/vendor/jquery/jquery.js  
inflating: templatemo_572_designer/vendor/jquery/jquery.min.js  
inflating: templatemo_572_designer/vendor/jquery/jquery.min.map  
inflating: templatemo_572_designer/vendor/jquery/jquery.slim.js  
inflating: templatemo_572_designer/vendor/jquery/jquery.slim.min.js  
inflating: templatemo_572_designer/vendor/jquery/jquery.slim.min.map  
[root@ip-172-31-32-240 ec2-user]# ls  
templatemo_572_designer templatemo_572_designer.zip  
[root@ip-172-31-32-240 ec2-user]# cd templatemo_572_designer/  
[root@ip-172-31-32-240 templatemo_572_designer]# ls  
about.html assets contact.html explore.html index.html trending.html vendor  
[root@ip-172-31-32-240 templatemo_572_designer]# mv * /var/www/html/  
mv: overwrite '/var/www/html/index.html'? y  
[root@ip-172-31-32-240 templatemo_572_designer]#
```

i-067aa27ef9ce7adf6 (haran)
PublicIPs: 16.171.154.21 PrivateIPs: 172.31.32.240

